

# A-MES

AUTOMOTIVE MES · SEOYON E-HWA

VOL 03

## Warehouse Management (WH)

Inbound · Inventory · Release · Locations

Document · L3 Detailed Design Specification

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Reference · Plant A " SAV Detroit, MI | Plant B " GEO Birmingham, AL

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WAREHOUSE MANAGEMENT

창고 관리 — PDA 바코드 · FIFO 출고 · 실시간 재고 · 입출고 이력

Full inbound/outbound warehouse management. **Key features:** SCM PO auto-receive (no PO = scan blocked), PDA barcode/QR scan for both inbound and outbound, real-time inventory by location, automatic min-stock alerts, mandatory FIFO picking (oldest LOT first), and full transaction audit trail. Integrates with PP (stock for MRP) and FG (finished goods inbound).

입고·출고 통합 창고 관리. **핵심 기능:** SCM 발주 자동 수신 (PO 없는 품목 스캔 차단), 입고/출고 모두 PDA 바코드·QR 스캔, 로케이션별 실시간 재고, 최소재고 자동 알림, FIFO 출고 강제 (입고일 오래된 LOT 우선), 입출고 전수 감사 이력. PP(MRP 재고)·FG(완제품 입고) 연계.

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2 (In · Out)

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Mandatory

QC TRIGGER

Auto on Inbound

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Alert MD-09/10

Click any screen card / icon / pill for L3 detailed spec · 각 화면 카드 클릭 시 L3 상세 설계서

WH-01 Inbound Schedule

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WH-03 Inventory Status

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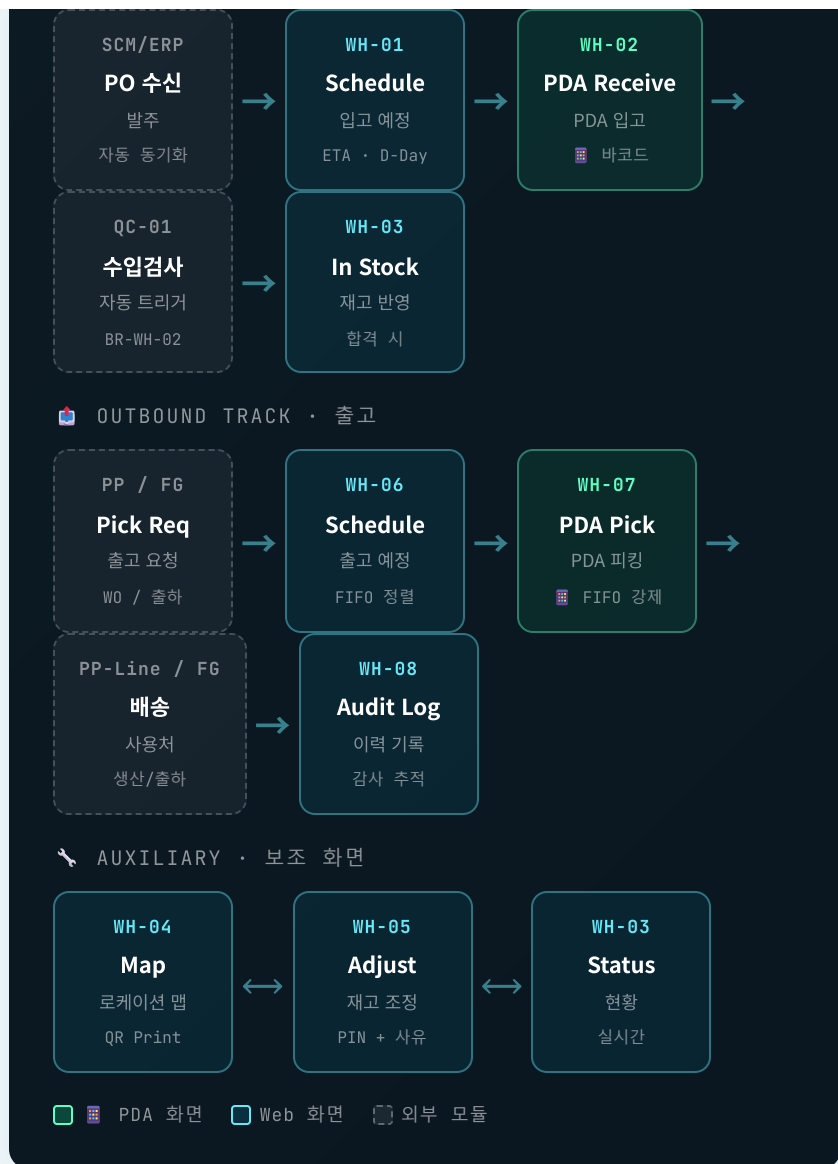
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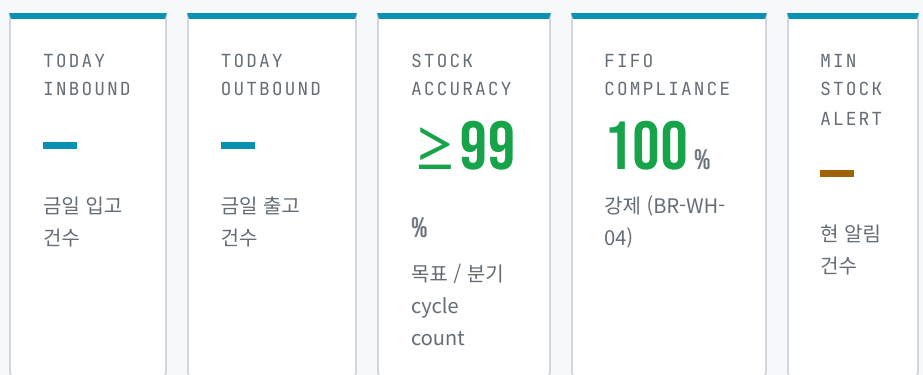
PROCESS FLOW · 화면 흐름

INBOUND & OUTBOUND TRACKS · 입출고 양방향 흐름

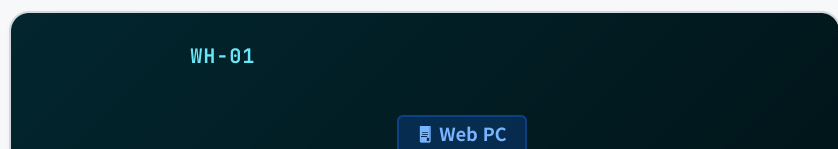
INBOUND TRACK · 입고



#### MODULE KPI · 핵심 지표



#### 8 SCREENS · 화면 설계







# Inbound Schedule

Phase 0 + 1st

L3 DETAIL →

입고 예정 — SCM  
PO 자동 수신 +  
ETA 관리

## What you'll see: SCM 발주 자동 수신 목록 + ETA D-Day



### 자동 계산 + 지연 빨강 강조

[Receive] 버튼 클릭 시 WH-02 PDA 입고 흐름으로 자동 분기

INPUT

**SCM PO  
Auto-Sync**

OUTPUT

**Schedule +  
WH-02  
trigger**

WATCH

**Overdue  
(red)**

UI

**Web Table  
+ Filter**

EN · DESCRIPTION

Displays all open purchase orders received from the SCM portal and allows warehouse staff to register expected inbound schedules. ETA D-Day is auto-calculated; overdue deliveries are highlighted in red. Clicking [Receive] triggers the WH-02 PDA inbound flow.

KO · 설명

SCM 포탈 수신 발주 기반 입고 예정 등록 화면. 입고 ETA D-Day 자동 계산, 지연 건 빨간색 강조. [Receive] 클릭 시 WH-02 PDA 입고 흐름으로 연계.



### Auto PO Sync

SCM 포탈 hourly polling.  
신규 PO 자동 수신



### ETA D-Day

자동 계산 + 카운트다운.  
D-3 노랑, D-0 빨강



### Overdue Highlight

납기 초과 행 빨간 배경 + 알림.  
SCM 자동 조회 알림



### → WH-02 Trigger

[Receive] 클릭 시 PDA 입고 진입.  
선택 행 자동 연계

WH-02 ·  CORE



## PDA

### Inbound Receiving

PDA 입고 처리 — 바코  
드 스캔 + 위치 지정

 PDA

 Core

 **L3 DETAIL →**



**What you'll see: PDA 스캔 화면 (품목 바코드 → 수량 숫자  
패드 → 선반 QR) 3단계**

PO 미등록 품목은 첫 스캔 즉시 차단 + QC 검사 자동 트리거

 INPUT

**Item  
Barcode +  
Qty + Loc  
QR**

 OUTPUT

**Stock IN  
(QC  
pending)**

 BLOCK

**PO 없는 품목**

 UI

**3-Step PDA  
Flow**

EN · DESCRIPTION

Warehouse staff scans item barcode via PDA, confirms quantity using the numeric keypad, then scans a shelf QR code to assign storage location. System validates barcode against PO. Mismatch is immediately blocked. Confirms stock entry, triggers QC-01 inspection, prints location label.

KO · 설명

창고 담당자가 PDA로 품목 바코드 스캔 → 숫자패드로 수량 입력 → 선반 QR 스캔으로 로케이션 지정. 바코드-PO 불일치 즉시 차단. 입고 확정 + QC-01 검사 자동 트리거 + 위치 라벨 출력.



#### 3-Step Scan

품목 바코드 → 수량 패드 → 위치 QR.  
1화면 1액션 원칙



#### PO Validation

미등록 바코드 즉시 차단.  
BR-WH-01



#### Auto QC Trigger

입고 즉시 QC-01 수입검사 발행.

합격 후 재고 반영



### Location Label

선반 QR 라벨 자동 인쇄.  
위치 부착용

관련 BR ▶

BR-WH-01

BR-WH-02

WH-03



## Inventory Status

Web PC

1st Dev

L3 DETAIL →

재고 현황 — 로케이  
선별 실시간

What you'll see: 품목/LOT/로케이션별 실시간 재고 그리드  
+ 최소재고 임박 노랑 강조

행 클릭 시 WH-08 입출고 이력 드릴다운

INPUT

All WH  
transactions

OUTPUT

Stock view  
+ Drill-  
down

ALERT

Min Stock  
(BR-WH-03)

UI

Searchable  
Web Grid

EN · DESCRIPTION

Real-time inventory query by item, LOT, location, or expiry. Min-stock items highlighted yellow with alert push to warehouse manager + procurement team. Drill-down per row to WH-08 transaction history. Excel export for cycle count.

KO · 설명

품목·LOT·로케이션·유효기간 실시간 검색. 최소재고 미만 행 노랑 강조 + 창고담당·구매팀 즉시 알림. 행 클릭 시 WH-08 입출고 이력 드릴다운. cycle count용 Excel 출력.



### Multi-Search

품목·LOT·위치·기간 동시 검색.  
필터 조합



### Min Stock Alert

MD-09/10 기준 임박 시 노랑·미만 시 빨강.  
+ 즉시 알림



### Drill-down

행 클릭 시 WH-08 이력.  
전체 트랜잭션 추적



### Excel Export

cycle count + 정기 점검용.  
7일 보관

관련 BR ▶

BR-WH-03

WH-04



## Location Map

로케이션 맵 — 시각화 + QR 출력

Web PC

2nd Dev

L3 DETAIL →

What you'll see: Zone × Row × Column 그리드 시각화  
(빈/부분/만재/초과 4색)

셀 클릭 시 위치별 품목 상세, QR 라벨 일괄 출력 지원

INPUT

MD-11  
Location  
Master

OUTPUT

Visual Map  
+ QR Labels

UI

Grid +  
Color  
Coded

ACTION

Reorganize  
(drag)

EN · DESCRIPTION

Visual warehouse layout: Zone → Row → Column → Level hierarchy. Each cell color-coded by capacity (Empty/Partial/Full/Over). QR label batch print for new locations. Drag to reorganize (supervisor PIN). Direct from MD-11 Location Master.

KO · 설명

창고 레이아웃 시각화: 구역 → 행 → 열 → 층 계층. 셀 용량별 4색(빈/부분/만재/초과) 표시. 신규 위치 QR 라벨 일괄 출력. 드래그로 재배치(라인장 PIN). MD-11 로케이션 마스터 직접 참조.



### 4-Color Capacity

Empty/Partial/Full/Over 즉시 인식.  
한눈 파악



### QR Batch Print

선반 라벨 일괄 출력.  
Zebra ZPL



### Drag Reorganize

레이아웃 변경 (PIN 필요).  
감사 기록



### MD-11 Sync

로케이션 마스터 실시간 미러.  
변경 즉시 반영

관련 BR ▶ 없음 (참조 화면)

WH-05



## Inventory Adjustment

Web PC

1st Dev

L3 DETAIL →

재고 조정 — 사유 코드 + 라인장 PIN

What you'll see: cycle count 차이 또는 손실/발견 시 사유



코드 + 수량 조정 입력 폼

모든 조정은 PIN 인증 + 7년 감사 로그 (BR-WH-05)



TRIGGER

Cycle count  
diff / Event



OUTPUT

Stock +/- +  
Audit Log



AUTH

Supervisor  
PIN



UI

Reason  
Code Form

EN · DESCRIPTION

Manual inventory adjustment with mandatory reason code (Damaged / Lost / Found / Count\_Diff / Other), supervisor PIN, and full audit trail. All changes recorded in WH-08 history with operator, approver, before/after qty. Audit log retained 7 years.

KO · 설명

수동 재고 조정 — 사유 코드 필수 (파손/분실/발견/실사차이/기타), 라인장 PIN 인증, 전수 감사 추적. 모든 변경이 WH-08 이력에 작업자·승인자·전후 수량과 함께 기록. 감사 로그 7년 보관.



### Reason Code Required

5개 사유 중 1개 강제 선택.

자유 텍스트 200자



### Supervisor PIN

라인장 인증 없이 조정 불가.  
감사 강화



### Before/After Snap

조정 전후 수량 자동 기록.  
롤백 가능



### 7-Year Audit

WH-08에 영구 기록.  
감사 요구사항

관련 BR ▶ 참조: BR-WH-03 (Min Stock)

WH-06



## Release Schedule

출고 예정 — 픽 리스  
트 + FIFO 자동 정렬

Web PC

1st Dev

L3 DETAIL →



What you'll see: PP WO 자재 요청 + FG 출하 요청 통합 목록 + FIFO 자동 추천 위치

[Pick] 버튼 클릭 시 WH-07 PDA 피킹 화면으로 자동 분기

INPUT

PP WO / FG  
Shipment

OUTPUT

Pick List +  
WH-07  
trigger

SORT

FIFO Auto

UI

Web Table  
+ Filter

EN · DESCRIPTION

Consolidates outbound requests from PP (material to production line) and FG (finished goods shipment). System auto-suggests pick locations following FIFO rule (oldest LOT first). [Pick] button transitions to WH-07 PDA picking flow with selected items.

KO · 설명

PP(생산라인 자재) + FG(완제품 출하) 출고 요청 통합. 시스템이 FIFO 규칙(입고일 오래된 LOT 우선)으로 피킹 위치 자동 추천. [Pick] 버튼 클릭 시 WH-07 PDA 피킹 화면으로 선택 항목 인계.



### PP + FG Merged

자재 + 출하 단일 큐.  
우선순위 통합



### FIFO Auto-Suggest

입고일 오래된 LOT 자동 추천.  
BR-WH-04 강제



### Due Date Priority

납기 임박 행 노랑·빨강.  
자동 정렬



### → WH-07 Trigger

[Pick] 클릭 시 PDA 피킹 진입.  
선택 인계

관련 BR ▶

BR-WH-04

WH-07 · CORE



## PDA Release Picking

PDA 출고 피킹 — FIFO  
강제 + 오폭목 차단

PDA

CORE

L3 DETAIL →



What you'll see: PDA 안내 화면 (시스템이 가장 오래된 LOT 위치로 안내 → 작업자 바코드 확인 → 수량)  
FIFO 위반 시도 또는 오폭목 스캔 즉시 차단 (BR-WH-04)

INPUT

Pick list  
(FIFO sorted)

OUTPUT

Stock OUT  
+ Manifest

BLOCK

Wrong item  
· FIFO violation

UI

PDA Guided Flow

EN · DESCRIPTION

PDA-guided FIFO picking for outbound. System suggests oldest stock location first. Operator scans item barcode to confirm correct pick; wrong-item scan immediately blocked.

KO · 설명

PDA 안내 기반 FIFO 출고 피킹. 시스템이 입고일 가장 오래된 LOT 위치 자동 제안. 작업자가 품목 바코드 스캔으로 정확성 확인 — 오폭목 즉시 차단. 다중 피킹 지원 + 통합 인보이스. 출고 확정 + 배송 전표 출력.

Multi-pick supported with consolidated manifest. Confirms outbound + prints delivery slip.



#### FIFO Guided

오래된 LOT 위치 자동 안내.  
현장 헤매기 0



#### Wrong-Item Block

오폐품 바코드 스캔 즉시 STOP.  
잘못된 출고 0



#### Multi-Pick

한 출고 요청 다중 LOT/위치 일괄.  
시간 단축



#### Manifest Print

배송 전표 자동 인쇄.  
PP/FG 인계

관련 BR ▶

BR-WH-04

WH-08



## Transaction History

Web PC

1st Dev

L3 DETAIL →

입출고 이력 — 전수  
감사 추적

What you'll see: 모든 입출고·조정 트랜잭션 통합 그리드 + 다중 필터 (날짜·품목·작업자·사유)

감사 대비 Excel 출력 + 영구 보관 (사유 코드별 색상)

INPUT

All WH events

OUTPUT

Audit Trail + Excel

RETAIN

7 years (감사)

UI

Multi-Filter Grid

EN · DESCRIPTION

Full audit trail of all inbound, outbound, and adjustment transactions: timestamp,

KO · 설명

모든 입고·출고·조정 트랜잭션 감사 기록: 일시·작업자·수량·위치·사유 코드·전후 스냅샷. 날짜/품목/작업



operator, quantity, location,  
reason code, before/after  
snapshot. Searchable by date  
range, item, operator, reason.  
Excel export for audit.  
Permanent retention 7 years.

자/사유 검색. 감사 대비 Excel 출력.  
7년 영구 보관.



### Multi-Filter

날짜·품목·작업자·사유 조합 검색.  
시점 추적 가능



### Excel Audit Export

감사 대비 전수 출력.  
서명 라인 포함



### Reason Color Code

사유별 행 색상 차별화.  
분석 용이



### Immutable

한 번 기록 후 변경 불가.  
7년 영구 보관

관련 BR ▶ 전 BR 이력 통합 보관

## ▼ LOCATION HIERARCHY · 로케이션 체계 (MD-11)

| Level      | Code Pattern | Example    | Description · 설명           |
|------------|--------------|------------|----------------------------|
| Zone · 구역  | [A-Z]        | A · B · C  | 창고 대구역 (자재 / 반제품 / 완제품 등). |
| Row · 행    | [1-99]       | A-12       | 구역 내 통로 번호.                |
| Column · 열 | [01-99]      | A-12-05    | 행 내 선반 위치.                 |
| Level · 층  | [01-09]      | A-12-05-03 | 선반 층(1층 = 최하단).            |

| Level   | Code Pattern            | Example    | Description · 설명     |
|---------|-------------------------|------------|----------------------|
| Full ID | {Z}-<br>{R}-<br>{C}-{L} | A-12-05-03 | QR 라벨 인쇄 양식. 4단계 계층. |

▼

ADJUSTMENT REASON CODES · 재고 조정 사유

| Code       | Name                  | KO    | When to Use · 사용 시점                        |
|------------|-----------------------|-------|--------------------------------------------|
| DAMAGED    | Damaged Stock         | 파손    | 이동/보관 중 파손 확인 시. 사진 첨부 권장. → 폐기 사유 코드와 별개. |
| LOST       | Lost Stock            | 분실    | cycle count 결과 시스템 재고보다 실재고가 적을 때.         |
| FOUND      | Found Stock           | 발견    | 예상 외 위치에서 발견되어 시스템에 반영.                    |
| COUNT_DIFF | Cycle Count Diff      | 실사 차이 | 정기 실사 후 시스템-실재고 차이 보정.                     |
| OTHER      | Other (text required) | 기타    | 위 4종에 해당하지 않는 경우. 자유 텍스트 200자 + PIN 필수.    |

▼

BUSINESS RULES · 업무 규칙 (WH · 4건)

| Rule ID  | Category        | Rule (EN)                                                                                                                       | 규칙 (KO)                           | Action |
|----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------|
| BR-WH-01 | Inbound Gate    | Items without a valid open PO cannot be received. Barcode scan is blocked immediately if no matching PO is found in the system. | PO 없는 품목 입고 불가. 미등록 바코드 스캔 즉시 차단. | Block  |
| BR-WH-02 | QC Auto-Trigger | All inbound items automatically trigger QC-01 incoming inspection. Inventory is updated only after QC                           | 입고 완료 즉시 QC-01 수입검사 자동 트리거.       | Auto   |

| Rule ID  | Category        | Rule (EN)                                                                                                                                             | 규칙 (KO)                                                                                                            | Action |
|----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------|
|          |                 | pass. QC fail<br>→ items<br>quarantined.                                                                                                              | 합격<br>후 재<br>고 반<br>영. 불<br>합격<br>시 격<br>리.                                                                        |        |
| BR-WH-03 | Min Stock Alert | When on-hand quantity falls below the minimum stock level (defined in MD-09/10), an alert is sent to the warehouse manager and procurement team.      | 현재<br>고 <<br>최소<br>재고<br>(MD-09/10<br>기준)<br>시 창<br>고담<br>당·구<br>매팀<br>즉시<br>알림.                                   | Alert  |
| BR-WH-04 | FIFO Picking    | All outbound picking must follow FIFO (First In First Out) based on inbound date. System auto-suggests oldest stock location. Wrong-LOT scan blocked. | 출고<br>피킹<br>입고<br>일 기<br>준<br>FIFO<br>자동<br>적용.<br>최고<br>령 재<br>고 위<br>치 자<br>동 제<br>안. 다<br>른<br>LOT<br>스캔<br>차단. | Auto   |

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VOL.3 · WH · WAREHOUSE MANAGEMENT · 입고 예정

WH-01INBOUND SCHEDULE

입고 예정

DEVICEWeb PC

PHASEPhase 0 + 1st Dev

ROLEWarehouse Staff · Manager

FREQDaily · 일 1회 이상

URL/wh/inbound/schedule

01Overview

화면 개요

Displays all open purchase orders received from the SCM portal and allows warehouse staff to register expected inbound schedules. ETA D-Day is auto-calculated; overdue deliveries are highlighted in red and trigger alerts to the procurement and warehouse manager. The [Receive] action launches the WH-02 PDA Inbound flow for the selected PO, while QC-01 incoming inspection is auto-queued upon confirmation.

SCM 포탈에서 수신한 모든 미입고 발주(PO)를 표시하고, 창고 담당자가 입고 예정 일정을 등록할 수 있는 화면입니다. ETA D-Day는 자동 계산되며, 지연 발주는 빨간색으로 강조되고 구매팀·창고 관리자에게 알림이 발송됩니다. [Receive] 클릭 시 선택한 PO에 대해 WH-02 PDA 입고 흐름이 시작되며, 확정 즉시 QC-01 수입검사가 자동 대기열에 등록됩니다.

|             |                                           |
|-------------|-------------------------------------------|
| SCREEN ID   | WH-01                                     |
| MODULE      | VOL.3 · Warehouse 창고 관리                   |
| LAYOUT      | Toolbar + KPI strip + Data table 툴바·KPI·표 |
| REFRESH     | Auto every 30 min + manual 30분 자동 + 수동    |
| PRIMARY KPI | Pending PO, Arriving Today, Overdue       |

## 02 Screen Mockup

### 화면 목업

## 02 Screen Mockup

화면 목업

## A-MES · Warehouse · Inbound Schedule

All Status

All Vendors

Search PO No. / Item...

연도 - 월 - 일

~

연도 - 월 - 일

Search

+ Register Schedule

Excel

|               |                   |               |                   |                   |
|---------------|-------------------|---------------|-------------------|-------------------|
| 12            | 3                 | 1             | 5                 | 3                 |
| Pending<br>PO | Arriving<br>Today | Overdue<br>지연 | This Week<br>이번 주 | Next Week<br>다음 주 |
| 입고 예정 건       | 금일 입고             |               |                   |                   |

| PO No.<br>발주 번호  | Vendor<br>공급사 | Item Code<br>품목코드 | Item Name<br>품목명                     | Order Qty<br>발주 수량 | Unit<br>단위 | ETA<br>입고 예정일 | D-Day |
|------------------|---------------|-------------------|--------------------------------------|--------------------|------------|---------------|-------|
| PO-2025-0628-003 | 삼성화학          | RM-PP-NAT-20      | PP Pellet Natural 20MFI<br>PP 펠렛 나투럴 | 500                | kg         | 2025-06-28    | D+3   |
| PO-2025-0701-001 | 현대원단          | FAB-GY-C03        | Fabric Roll Grey GY-C03<br>원단 롤 회색   | 20                 | roll       | 2025-07-01    | Today |
| PO-2025-0705-002 | KCC도료         | PNT-BK-NH578      | Black Paint NH578<br>블랙 도료 NH578     | 50                 | L          | 2025-07-05    | D-4   |
| PO-2025-0708-004 | 현대원단          | FAB-BK-C01        | Fabric Roll Black BK-C01             | 10                 | roll       | 2025-07-08    | D-7   |

| PO No.<br>발주<br>번호           | Vendor<br>공급사 | Item Code<br>품목코<br>드    | Item Name<br>품목명                              | Order Qty<br>발주 수<br>량 | Unit<br>단위 | ETA<br>입고 예<br>정일 | D-Day |
|------------------------------|---------------|--------------------------|-----------------------------------------------|------------------------|------------|-------------------|-------|
|                              |               |                          | 원단 롤<br>블랙                                    |                        |            |                   |       |
| PO-<br>2025-<br>0710-<br>005 | 삼성화<br>학      | RM-<br>PP-<br>BLK-<br>20 | PP<br>Pellet<br>Black<br>20MFI<br>PP 펠렛<br>블랙 | 300                    | kg         | 2025-<br>07-10    | D-9   |

**i Row color legend · 행 색상** — Red row: Overdue PO · 지연 / Green row: Arriving today · 금일 도착 / White: Scheduled · 예정. D-Day 색상: **D+N (red)** · 지연, **Today (green)** · 금일, **D-N (default)** · 예정.

## 03 UI Component Specification

UI 컴포넌트 명세

### 3.1 Toolbar Components · 툴바 컴포넌트

| ID     | COMPONENT<br>컴포넌트          | TYPE<br>타입 | REQ<br>필수 | VALIDATION / BEHAVIOR<br>검증·동작                                                                       | DEFAULT<br>기본값 |
|--------|----------------------------|------------|-----------|------------------------------------------------------------------------------------------------------|----------------|
| CMP-01 | Status Filter<br>상태 필터     | SELECT     | N         | Options: All / Pending / Arriving Today / Overdue / Completed . Changes trigger table reload.        | All Status     |
| CMP-02 | Vendor Filter<br>공급사 필터    | SELECT     | N         | Loaded from md_vendor where is_active=1 . Multi-select disabled in v1.                               | All Vendors    |
| CMP-03 | Search Box<br>검색창          | TEXT       | N         | Min 2 chars. Searches po_no and item_code/name with LIKE %query% . Debounced 300ms.                  | (empty)        |
| CMP-04 | Date Range<br>기간 선택        | DATE × 2   | N         | From ≤ To. Max range 90 days. Filter on eta_date .                                                   | Today ~ +14d   |
| CMP-05 | Search Button<br>검색 버튼     | BUTTON     | N         | Triggers GET /api/wh/inbound/schedule with all filter params. Enter key in any filter also triggers. | —              |
| CMP-06 | Register Schedule<br>예정 등록 | BUTTON     | N         | Opens modal for manual schedule entry (vendor-driven PO without SCM sync). Requires WH.Manage role.  | —              |

| ID     | COMPONENT<br>컴포넌트     | TYPE<br>타입 | REQ<br>필수 | VALIDATION / BEHAVIOR<br>검증·동작                                                                              | DEFAULT<br>기본값 |
|--------|-----------------------|------------|-----------|-------------------------------------------------------------------------------------------------------------|----------------|
| CMP-07 | Excel Export<br>엑셀 출력 | BUTTON     | N         | Exports current filtered result as<br>.xlsx . Max 10,000 rows.<br>Filename:<br>WH01_Inbound_YYYYMMDD.xlsx . | -              |

3.2 KPI Strip · KPI 영역

| ID     | METRIC<br>지표             | TYPE  | CALCULATION<br>산출 로직                                                           | REFRESH |
|--------|--------------------------|-------|--------------------------------------------------------------------------------|---------|
| KPI-01 | Pending PO<br>입고 예정<br>건 | COUNT | Count of po_header where status IN ('Scheduled', 'Arriving', 'Overdue') .      | 30 min  |
| KPI-02 | Arriving Today<br>금일 입고  | COUNT | Count where eta_date = CURRENT_DATE . Color: #22c55e .                         | 30 min  |
| KPI-03 | Overdue<br>지연            | COUNT | Count where eta_date < CURRENT_DATE AND status ≠ 'Received' . Color: #ef4444 . | 30 min  |
| KPI-04 | This Week<br>이번 주        | COUNT | Count where eta_date within current ISO week. Excludes Today.                  | 30 min  |
| KPI-05 | Next Week<br>다음 주        | COUNT | Count where eta_date within next ISO week.                                     | 30 min  |

3.3 Data Table Columns · 데이터 표 컬럼

| ID     | COLUMN<br>컬럼      | SOURCE<br>소스      | FORMAT / RULE<br>표시·규칙                                                 | SORT |
|--------|-------------------|-------------------|------------------------------------------------------------------------|------|
| COL-01 | PO No.<br>발주번호    | po_header.po_no   | Format: PO-YYYY-MMDD-NNN . Overdue: red bold. Click → PO detail popup. | DESC |
| COL-02 | Vendor<br>공급사     | md_vendor.name    | Display vendor short name. Tooltip = full legal name.                  | ASC  |
| COL-03 | Item Code<br>품목코드 | md_item.item_code | Monospace font. Click → item master popup.                             | ASC  |
|        |                   |                   |                                                                        |      |

| ID     | COLUMN<br>컬럼          | SOURCE<br>소스            | FORMAT /<br>RULE<br>표시·규칙                                                                               | SORT     |
|--------|-----------------------|-------------------------|---------------------------------------------------------------------------------------------------------|----------|
| COL-04 | Item<br>Name<br>품목명   | md_item.item_name_en/ko | EN main + KO<br>subtext<br>(smaller,<br>muted).                                                         | —        |
| COL-05 | Order<br>Qty<br>발주 수량 | po_header.order_qty     | Right-align.<br>Thousands<br>separator.<br>Decimal: 0–3<br>digits per unit.                             | —        |
| COL-06 | Unit<br>단위            | md_item.unit            | kg / roll / L /<br>EA.                                                                                  | —        |
| COL-07 | ETA<br>입고 예정일         | po_header.eta_date      | Format: YYYY-<br>MM-DD .<br>Editable inline<br>by Manager<br>role.                                      | ASC<br>★ |
| COL-08 | D-Day<br>잔여일          | Calculated              | eta_date -<br>CURRENT_DATE .<br>Show D+N<br>(red) / Today<br>(green) / D-N .                            | —        |
| COL-09 | Status<br>상태          | po_header.status        | Badge:<br>Scheduled<br>(blue) /<br>Arriving<br>(yellow) /<br>Overdue<br>(red) /<br>Received<br>(green). | —        |
| COL-10 | Action<br>액션          | Button                  | [Receive] if<br>status in<br>(Arriving /<br>Overdue /<br>Scheduled<br>within 1 day) ·<br>else [Detail]. | —        |

## 04 Event & Action Flow

이벤트 · 액션 흐름

1

**SCHEDULED** scm.sync.cron @ \*/30min

System fetches new/updated POs from SCM portal every 30 minutes via SAP B1 SLD adapter.

시스템이 SCM 포탈에서 신규/갱신 PO를 30분마다 SAP B1 SLD 어댑터로 수신합니다.



2

**EFFECT** UPSERT into `po_header` · UPDATE `last_synced_at` · trigger overdue re-evaluation.

**USER** `page.load` · WH-01 진입

Warehouse staff opens the screen. System auto-applies default filter (Today ~ +14d, All Status).

창고 담당자가 화면에 진입하면 기본 필터(오늘~+14일, 전체 상태)가 자동 적용됩니다.

**API** `GET /api/wh/inbound/schedule?`  
`from=...&to=...&status=all` → KPI strip + table populated.

3

**USER** `click` · [ Search]

User adjusts filter (status / vendor / date range / search box) and clicks Search. Same effect when pressing Enter in any filter field.

필터(상태/공급사/기간/검색) 변경 후 [검색] 클릭 또는 Enter 입력 시 동일 동작.

**VALIDATE** `date_from ≤ date_to`, range ≤ 90 days, search ≥ 2 chars → re-query & redraw table.

4

**USER** `click` · [ Receive]

User clicks Receive on a row. System checks RBAC WH.Receive and PO status validity, then launches WH-02 PDA flow with PO context.

[Receive] 클릭 시 RBAC 권한·PO 상태 검증 후 WH-02 PDA 입고 흐름으로 PO 정보와 함께 전환됩니다.

**REDIRECT** `POST /api/wh/inbound/start` → returns `session_id` → redirect `/wh/pda/inbound?po=...&sid=...`.

5

**SYSTEM** `job.overdue.check @ 09:00 daily`

Daily 09:00 KST job re-evaluates ETA. POs past ETA without receipt become Overdue → alert sent to procurement & WH manager via in-app + email.

매일 09:00 KST 배치가 ETA를 재평가하여 미입고 PO를 지연 상태로 전환 + 구매팀·창고관리자에게 알림 발송.

**EFFECT** `UPDATE po_header SET status='Overdue'` · INSERT into `sys_alert` · email via SYS-04.

6

**USER** `click` · [ Register Schedule]

Manager opens modal to manually register an inbound schedule for non-SCM POs (e.g. local consumables, sample shipments).

SCM 외 발주(소모성·샘플 등)에 대해 매니저가 수동으로 입고 예정을 등록할 수 있는 모달을 엽니다.



**REQUIRE** Role = `WH.Manage` · fields: Vendor, Item, Qty, ETA,  
Note → `POST /api/wh/inbound/manual` .

**SYSTEM** `return` · from `WH-02 success`

After WH-02 PDA inbound completion, user returns to WH-01. Received PO row disappears (or shows Received badge depending on filter).

WH-02 PDA 입고 완료 후 WH-01로 복귀하면 입고 완료된 PO 행이 사라지거나 (필터에 따라) [Received] 상태 배지로 표시됩니다.

**DOWNSTREAM** QC-01 incoming inspection auto-queued · min-stock alert re-evaluated · WH-08 transaction log written.

## 05

### Exception & Error Handling

예외 · 에러 처리

**E-WH01-001**

#### SCM Sync Failure

SCM 동기화 실패

|        |                                                                                                                          |
|--------|--------------------------------------------------------------------------------------------------------------------------|
| CAUSE  | SAP B1 SLD endpoint unreachable / timeout > 30s.<br>SAP B1 SLD 연결 실패 · 30초 초과.                                           |
| DETECT | Last <code>last_synced_at</code> > 60 min ago.                                                                           |
| ACTION | Show yellow banner at top: "Last sync: HH:MM. Connection issue."<br>Retry button available. Cached data still displayed. |
| NOTIFY | SYS admin via in-app alert + email after 3 consecutive failures.                                                         |

⚠ SCM SYNC DELAYED — DATA MAY BE STALE

**E-WH01-002**

#### Receive Without Permission

권한 없는 입고 시도

|        |                                                                                                                           |
|--------|---------------------------------------------------------------------------------------------------------------------------|
| CAUSE  | User clicks [Receive] without <code>WH.Receive</code> role.<br>권한 없는 사용자가 입고 시도.                                          |
| DETECT | Backend RBAC check on <code>POST /api/wh/inbound/start</code> .                                                           |
| ACTION | Block API call. Return <code>403 Forbidden</code> . Modal: "You don't have permission to receive. Contact your manager."  |
| LOG    | Record attempt in <code>sys_audit</code> with <code>user_id</code> , <code>timestamp</code> , <code>attempted_po</code> . |

🚫 ACCESS DENIED — ROLE WH.RECEIVE REQUIRED

**E-WH01-003**

#### PO Already Received

이미 입고된 PO

|                                        |                                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------|
| CAUSE                                  | Another user already completed WH-02 for this PO (race condition).<br>동시 작업으로 이미 다른 사용자가 입고 완료.         |
| DETECT                                 | On [Receive] click: <code>SELECT status FROM po_header</code> → status = 'Received'.                    |
| ACTION                                 | Refuse navigation to WH-02. Show toast: "This PO has been received by [user] at [time]." Refresh table. |
| RECOVERY                               | User views WH-08 Transaction History for confirmation.                                                  |
| ⚠ ALREADY RECEIVED — SEE WH-08 HISTORY |                                                                                                         |

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| E-WH01-004 Invalid Date Range<br>잘못된 기간 입력 |                                                                                                                   |
| CAUSE                                      | <code>date_from &gt; date_to</code> or range > 90 days.<br>시작일 > 종료일 또는 기간 90일 초과.                                |
| DETECT                                     | Client-side validation before API call.                                                                           |
| ACTION                                     | Highlight date inputs with red border. Inline error: "Range must be valid and ≤ 90 days." Search button disabled. |
| RECOVERY                                   | User corrects dates → validation passes → button re-enabled.                                                      |
| ⚠ INVALID RANGE — MAX 90 DAYS              |                                                                                                                   |

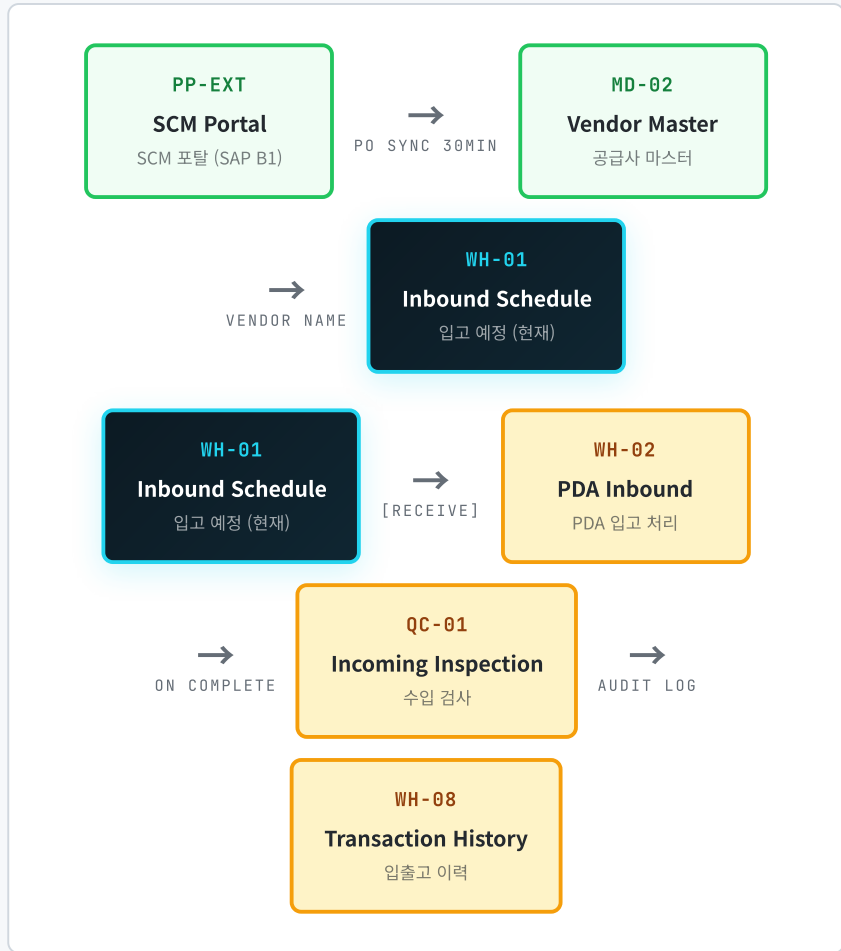
|                                               |                                                                                               |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|
| E-WH01-005 Excel Export Overflow<br>엑셀 출력력 초과 |                                                                                               |
| CAUSE                                         | Filtered result > 10,000 rows.<br>필터 결과가 1만 행 초과.                                             |
| DETECT                                        | Server-side count check before file generation.                                               |
| ACTION                                        | Reject export. Modal: "Result exceeds 10,000 rows. Narrow your filter or use Reports module." |
| RECOVERY                                      | User reduces date range or applies vendor filter, then retries.                               |
| ⚠ TOO MANY ROWS — NARROW FILTER               |                                                                                               |

|                                         |                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| E-WH01-006 Empty Result Set<br>조회 결과 없음 |                                                                                                                               |
| CAUSE                                   | No PO matches current filter.<br>필터 조건에 맞는 발주 없음.                                                                             |
| DETECT                                  | API returns empty array.                                                                                                      |
| ACTION                                  | Show empty state illustration + message: "No inbound schedules match your filter." KPI strip shows zeros but remains visible. |
| RECOVERY                                | Reset filter button shown.                                                                                                    |
| i NO RESULTS — TRY CLEARING FILTER      |                                                                                                                               |

## 06

## Linked Screens

선행 · 후행 연계 화면



## 07

## Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                                                    | ACTION |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-01 | Items without a valid open PO cannot be received. Barcode scan is blocked immediately if no matching PO is found in the system.<br>유효한 미입고 PO가 없는 품목은 입고할 수 없으며, 일치하는 PO가 없는 바코드 스캔은 즉시 차단됩니다. | Block  |
| BR-WH-02 | All inbound items automatically trigger QC-01 incoming inspection. Inventory is updated only after QC pass.<br>모든 입고 품목은 QC-01 수입검사가 자동 트리거되며, 검사 합격 후에만 재고에 반영됩니다.                            | Auto   |
| BR-WH-03 | When on-hand quantity falls below minimum stock level (MD-09/10), an alert is sent to the warehouse manager and procurement team.                                                              | Alert  |

| RULE ID | DESCRIPTION                                          | ACTION |
|---------|------------------------------------------------------|--------|
|         | 현재고가 최소재고(MD-09/10) 미만이 되면 창고 관리자·<br>구매팀에 즉시 알림 발송. |        |

✳ **Note · 비고** — BR-WH-04 (FIFO) does not apply to inbound flow; it governs outbound picking (WH-07). For full WH business rules, refer to **VOL.3 WH module — Business Rules section.**

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PDA INBOUND RECEIVING

PDA 입고

DEVICE

PDA

PHASE

1st Dev

ROLE

Warehouse Staff (WH.Receive)

FREQ

Per inbound

입고 시

URL

/wh/pda/inbound

01

Overview

화면 개요

PDA-driven receiving workflow consisting of three sequential steps: (1) Scan item barcode against the open PO, (2) Confirm quantity via numeric keypad, and (3) Scan shelf QR code to assign storage location. The system validates each step in real time — barcode mismatches against the PO item are immediately blocked, and the worker cannot proceed until correction. Upon confirmation, inventory is incremented, a location label is printed, and the QC-01 incoming inspection queue is auto-triggered.

PDA 기반 3단계 입고 처리: (1) 품목 바코드 스캔으로 PO와 대조, (2) 숫자패드로 수량 확인, (3) 선반 QR 스캔으로 로케이션 지정. 각 단계에서 실시간 검증되며, PO 품목과 불일치 시 즉시 차단되어 수정 전까지 진행 불가. 확정 시 재고 증가·위치 라벨 출력·QC-01 수입검사 큐 자동 등록이 이루어집니다.

|           |                                         |
|-----------|-----------------------------------------|
| SCREEN ID | WH-02                                   |
| ENTRY     | From WH-01 [Receive] button WH-01 입고 클릭 |
| LAYOUT    | 3-step wizard · full-screen 3단계 위저드     |
| HARDWARE  | Zebra TC22 · 1D/2D scanner              |
| NETWORK   | WiFi · offline buffer 4hr 오프라인 4시간 버퍼   |
| TABLES    | wh_inbound , wh_inventory , po_header   |

## 02 Screen Mockup

화면 목록 — PDA 3단계

### ● WH-02 · PDA INBOUND · 3-STEP WIZARD

#### ① SCAN ITEM · 품목 스캔

WH-02 · Inbound 09:12

### Inbound Receiving

입고 처리 · PO-2025-0701-001



#### Scan Item Barcode

품목 바코드 스캔

PO NO.

PO-2025-0701-001

EXPECTED · 예정 품목

FAB-GY-C03

VENDOR

현대원단

#### ② CONFIRM QTY · 수량 입력

WH-02 · Inbound 09:13

### Confirm Quantity

수량 확인 · FAB-GY-C03 ✓

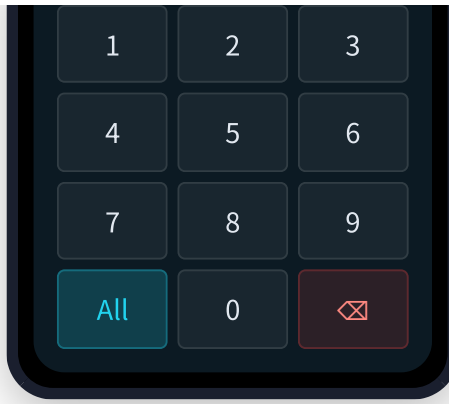
ITEM VERIFIED ✓

FAB-GY-C03 · Grey Fabric

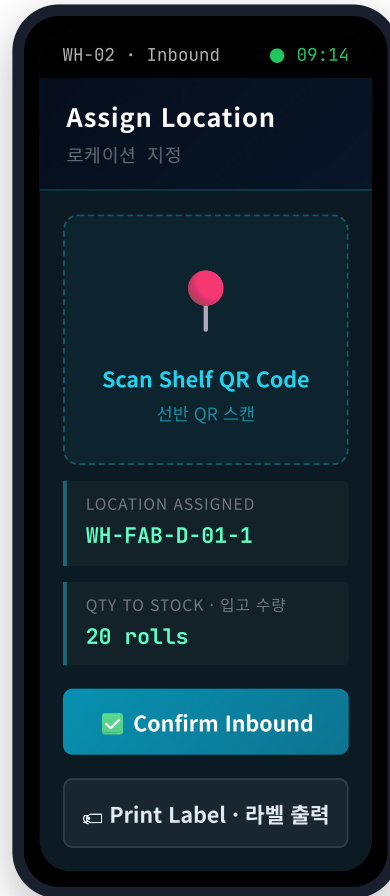
EXPECTED · 예정 수량

20 rolls

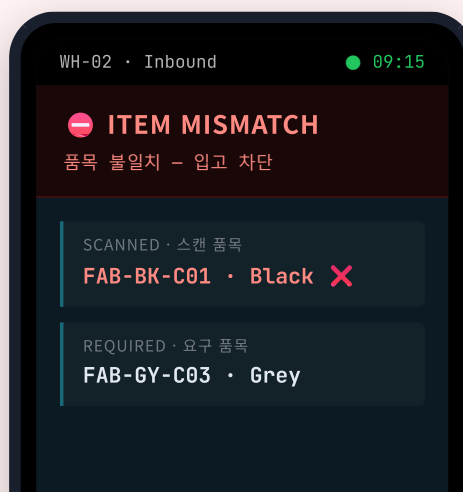
20



### ③ ASSIGN LOCATION · 위치 지정



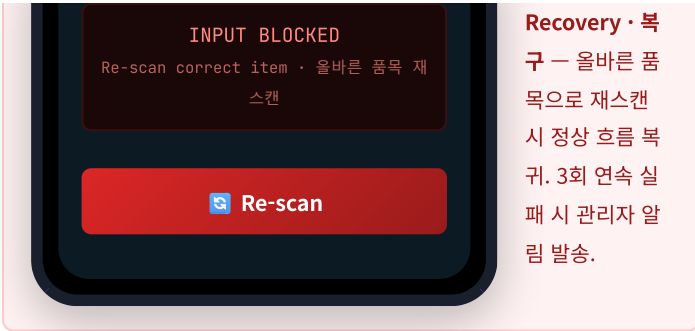
### ⊖ Error Case · 에러 케이스 — Item Mismatch · 품목 불일치



**Trigger · 발생 시점** — Step ①  
품목 바코드가  
PO 품목과 불일치 시.

**Action · 동작**  
— 입력 즉시 차단 · 빨간 화면 + 진동 + 비프음  
알림 · 재스캔 버튼만 활성화.





# 03 UI Component Specification

UI 컴포넌트 명세

## 3.1 Step ① - Item Scan · 품목 스캔

| ID     | COMPONENT<br>컴포넌트          | TYPE   | REQ | VALIDATION /<br>BEHAVIOR<br>검증·동작                                                                                            | ON<br>SUCCESS |
|--------|----------------------------|--------|-----|------------------------------------------------------------------------------------------------------------------------------|---------------|
| CMP-01 | Barcode Scanner<br>바코드 스캐너 | SCAN   | Y   | Reads 1D/2D barcode (Code128, QR). Match against <code>po_header.item_code</code> . Mismatch → block.                        | → Step ②      |
| CMP-02 | PO Display<br>PO 표시        | LABEL  | -   | Read-only. Shows <code>po_no</code> , expected item code, vendor name from launch context.                                   | -             |
| CMP-03 | Manual Entry<br>수동 입력      | BUTTON | N   | Fallback if barcode damaged. Opens keyboard input. Same validation as scanner. Logs <code>manual_entry=true</code> in audit. | → Step ②      |

## 3.2 Step ② - Quantity Confirmation · 수량 확인

| ID     | COMPONENT<br>컴포넌트      | TYPE   | REQ | VALIDATION /<br>BEHAVIOR                                                       | ON<br>SUCCESS |
|--------|------------------------|--------|-----|--------------------------------------------------------------------------------|---------------|
| CMP-04 | Numeric Keypad<br>숫자패드 | NUMPAD | Y   | 0-9, decimal point (per unit precision), backspace. Max 6 digits + 3 decimals. | → Step ③      |

| ID     | COMPONENT<br>컴포넌트                         | TYPE    | REQ | VALIDATION<br>/ BEHAVIOR                                                                                         | ON<br>SUCCESS |
|--------|-------------------------------------------|---------|-----|------------------------------------------------------------------------------------------------------------------|---------------|
| CMP-05 | <b>[All] Button</b><br>전량 버튼              | BUTTON  | N   | Auto-fills with full expected quantity from PO. Shortcut for common case.                                        | → Step ③      |
| CMP-06 | <b>Qty Display</b><br>수량 표시               | DISPLAY | —   | Large monospace digits. Color: cyan if within tolerance $\pm 2\%$ of expected, yellow $\pm 10\%$ , red $>10\%$ . | —             |
| CMP-07 | <b>Partial Inbound Toggle</b><br>부분 입고 토글 | SWITCH  | N   | If qty < expected, this auto-enables. PO status becomes <b>Partial</b> , remaining qty stays open.               | —             |

### 3.3 Step ③ — Location Assignment · 위치 지정

| ID     | COMPONENT                            | TYPE   | REQ | VALIDATION / BEHAVIOR                                                                                               | ON<br>SUCCESS |
|--------|--------------------------------------|--------|-----|---------------------------------------------------------------------------------------------------------------------|---------------|
| CMP-08 | <b>Shelf QR Scanner</b><br>선반 QR 스캐너 | SCAN   | Y   | Reads location QR. Format: <b>[Warehouse]-[Zone]-[Row]-[Col]-[Level]</b> . Check capacity from <b>md_location</b> . | → Confirm     |
| CMP-09 | <b>Suggested Location</b><br>권장 위치   | LABEL  | —   | System suggests location based on item type + nearest available cell with capacity. Worker can override.            | —             |
| CMP-10 | <b>[Confirm Inbound]</b><br>입고 확정    | BUTTON | Y   | Commits transaction. Increments <b>wh_inventory.on_hand_qty</b> . Inserts <b>wh_inbound</b> . Triggers QC-01.       | → Print       |
| CMP-11 | <b>[Print Label]</b><br>라벨 출력        | BUTTON | N   | Prints location label via Zebra ZD230 over Bluetooth. Includes item, qty, location, inbound date.                   | —             |

1

**USER** launch · from WH-01 [Receive]

User clicks [Receive] on a PO row in WH-01. PDA launches WH-02 with PO context. Step ① is shown.

WH-01에서 [Receive] 클릭 시 WH-02가 PO 정보와 함께 실행되며 Step ① 화면 표시.

**CONTEXT** `session_id`, `po_no`, `expected_item` loaded into PDA local state.

2

**PDA SCAN** trigger · item barcode

Worker pulls trigger to scan item barcode. System matches against PO expected item.

작업자가 트리거를 당겨 품목 바코드 스캔. 시스템이 PO 예정 품목과 대조.

**VALIDATE** `scanned_code == po.item_code` → success (green) · mismatch → block (E-WH02-001).

3

**USER** tap · numpad / [All]

Worker enters received quantity via numpad or taps [All] to auto-fill expected quantity. Display shows qty with tolerance color coding.

숫자패드 입력 또는 [All] 버튼으로 수량 입력. 디스플레이는 허용오차 색상 표시.

**DETECT** `qty < expected` → auto-enable Partial Inbound mode.  
`qty > expected` → warn (E-WH02-003).

4

**PDA SCAN** trigger · shelf QR

Worker scans the shelf QR code at the target storage location. System validates capacity vs incoming qty.

대상 선반의 QR 코드를 스캔. 시스템이 용량 대비 입고량 검증.

**VALIDATE** `location.capacity - location.on_hand ≥ qty` → OK · else block (E-WH02-004).

5

**USER** tap · [Confirm Inbound]

Worker confirms. Backend commits transaction atomically: insert inbound, update inventory, update PO status, queue QC-01.

작업자 확정. 백엔드가 트랜잭션을 원자적으로 커밋: 입고 INSERT, 재고 UPDATE, PO 상태 변경, QC-01 큐 등록.

**API** `POST /api/wh/inbound/confirm` · `BEGIN TRAN ... COMMIT` · returns `inbound_id`.

6

**SYSTEM** print · location label

Optionally prints a location label via Bluetooth-paired Zebra ZD230 printer. Worker affixes label to the lot.

블루투스 페어링된 Zebra ZD230 프린터로 위치 라벨 출력(선택). 작업자가 로트에 부착.

**OUTPUT** ZPL command via Bluetooth · label contains item, qty, location, date, inbound\_id barcode.

**SYSTEM** complete · return to WH-01

PDA returns to WH-01. Received PO is no longer in pending list. QC-01 inspector receives notification.

PDA가 WH-01로 복귀. 처리된 PO는 미입고 목록에서 제외. QC-01 검사자에게 알림.

**DOWNSTREAM** QC-01 queue · WH-03 inventory updated · WH-08 transaction log · min-stock check.

## 05

## Exception & Error Handling

예외 · 에러 처리

E-WH02-001

**Item Barcode Mismatch**

품목 바코드 불일치

|          |                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------|
| CAUSE    | Scanned item code $\neq$ PO expected item.<br>스캔 품목이 PO 예정 품목과 다름.                                       |
| DETECT   | Step ① real-time comparison.                                                                             |
| ACTION   | Red full-screen warning + vibration + beep. Re-scan button only. 3 consecutive failures → manager alert. |
| RECOVERY | Re-scan correct item to resume.                                                                          |

ITEM MISMATCH — INPUT BLOCKED

E-WH02-002

**PO Not Found / Expired**

PO 없음 · 만료

|          |                                                                                                |
|----------|------------------------------------------------------------------------------------------------|
| CAUSE    | PO context invalid: cancelled, already received, or session timeout.<br>PO 취소 · 이미 입고 · 세션 만료. |
| DETECT   | Server check on Step ① scan.                                                                   |
| ACTION   | Block scan. Modal: "PO no longer valid. Return to WH-01." Auto-redirect after 5 sec.           |
| RECOVERY | Re-select PO from WH-01.                                                                       |

PO INVALID — RETURN TO LIST

E-WH02-003

### Qty Exceeds PO

발주량 초과

|          |                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------|
| CAUSE    | Entered qty > expected by >5%.<br>입력 수량이 발주 수량 5% 초과.                                                     |
| DETECT   | Step ② numeric validation.                                                                                |
| ACTION   | Yellow display. Confirmation popup: "Receive QTY (over by N%)? Comment required." Free-text reason input. |
| RECOVERY | User confirms with reason, or corrects qty.                                                               |

⚠ OVER-RECEIPT — REASON REQUIRED

E-WH02-004

### Location Capacity Exceeded

로케이션 용량 초과

|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| CAUSE    | Target location's free capacity < incoming qty.<br>대상 위치 잔여 용량 부족.    |
| DETECT   | Step ③ on QR scan.                                                    |
| ACTION   | Block confirmation. Show free capacity. Suggest alternative location. |
| RECOVERY | Scan different location, or split into multiple locations.            |

🚫 LOCATION FULL — SCAN ANOTHER

E-WH02-005

### Network Disconnected

네트워크 단절

|          |                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------|
| CAUSE    | WiFi signal lost or backend unreachable.<br>WiFi 단절 또는 백엔드 응답 없음.                                         |
| DETECT   | PDA heartbeat check every 10s.                                                                            |
| ACTION   | Switch to offline mode. Queue transactions locally (up to 4hr).<br>Yellow banner. Auto-sync on reconnect. |
| RECOVERY | Reconnect to WiFi → queued transactions sync in FIFO order.                                               |

⚠ OFFLINE — QUEUE MODE ACTIVE

E-WH02-006

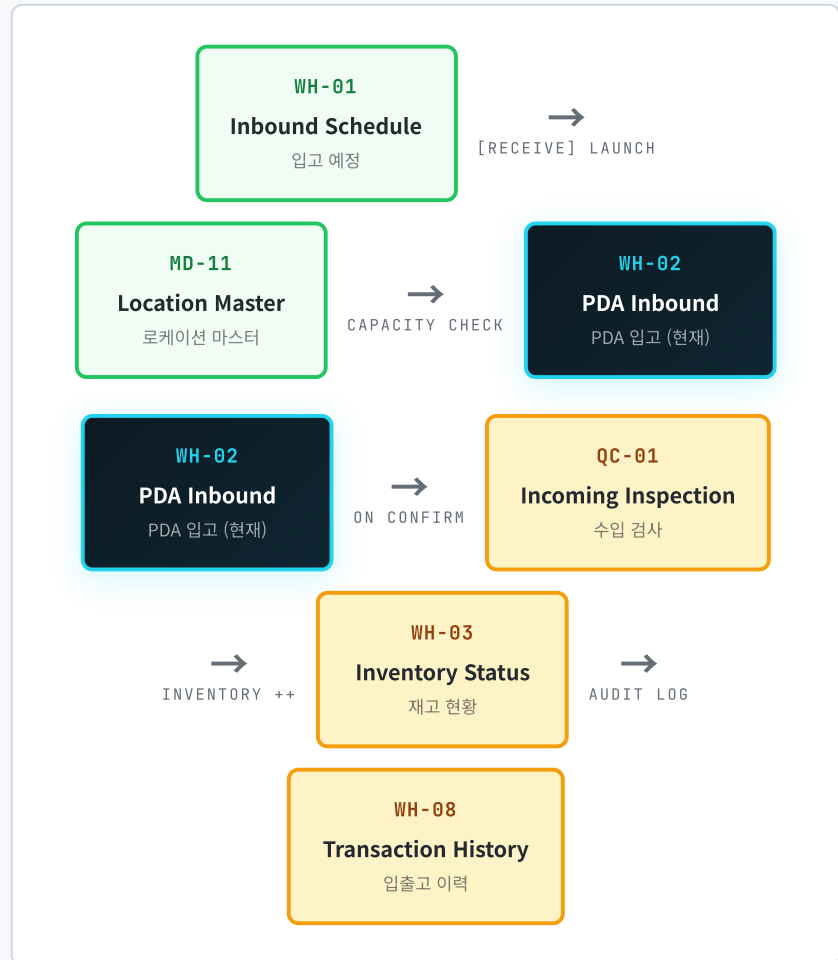
### Printer Not Ready

프린터 미응답

|          |                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------|
| CAUSE    | Bluetooth printer disconnected or out of paper.<br>블루투스 단절·용지 없음.                                          |
| DETECT   | Print request timeout 8s.                                                                                  |
| ACTION   | Inbound transaction NOT rolled back. Show: "Label print failed. Retry?" Inventory commit already complete. |
| RECOVERY | Re-print from WH-08 transaction history.                                                                   |

## 06 Linked Screens

선행 · 후행 연계 화면



## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                         | ACTION |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-01 | <p>Items without a valid open PO cannot be received. Barcode scan blocked immediately if no matching PO is found.</p> <p>유효한 미입고 PO 없는 품목 입고 불가. 미일치 바코드 즉시 차단.</p> | Block  |

| RULE ID  | DESCRIPTION                                                                                                                                       | ACTION |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-02 | All inbound items automatically trigger QC-01 incoming inspection. Inventory updated only after QC pass.<br>모든 입고 품목 QC-01 자동 트리거. 검사 합격 후 재고 반영. | Auto   |
| BR-WH-03 | When on-hand qty falls below minimum stock level (MD-09/10), an alert is sent to manager and procurement.<br>최소재고 미달 시 창고관리자·구매팀 알림 발송.           | Alert  |



**Note · 비고** — WH-02 is the only entry point for inbound transactions. Manual inventory entry must use WH-05 (Adjustment), which requires supervisor approval. WH-02는 입고 거래의 유일한 진입점이며, 수동 입력은 WH-05 (조정)로 처리하고 관리자 승인이 필요합니다.

A-MES

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INJ Injection

IMG Wrapping

PNT Painting

QC Quality

FG Finished Goods

MNT Maintenance

RPT

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VOL.3

WH Warehouse

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VOL.3

WH

WAREHOUSE MANAGEMENT

재고 현황

WH-03

INVENTORY STATUS

재고 현황

DEVICEWeb PC

PHASE1st Dev

ROLEAll warehouse roles

FREQReal-time · 실시간

URL/wh/inventory

01

Overview

화면 개요

Real-time inventory dashboard showing stock by item, material type (Raw Material / Fabric / Paint), and location. Items below the minimum stock level (from MD-09 · Min/Max Stock Master) are highlighted in red and surface in the KPI strip. The screen supports Excel export of the filtered view and shows daily snapshot delta vs. the previous day's closing inventory, so staff can quickly spot abnormal consumption or replenishment needs.

품목·유형(원자재/원단/도료)·로케이션별 실시간 재고 현황 대시보드입니다. 최소재고( MD-09 · 최소·최대재고 마스터) 미달 항목은 빨간색으로 강조되고 KPI에 별도 집계됩니다. 필터링된 결과의 엑셀 출력을 지원하며, 전일 마감 재고 대비 증감을 표시하여 비정상 소비·보충 필요 항목을 빠르게 식별할 수 있습니다.

|           |                                                  |
|-----------|--------------------------------------------------|
| SCREEN ID | WH-03                                            |
| MODULE    | VOL.3 · Warehouse 창고 관리                          |
| LAYOUT    | Filter bar + KPI + Data table 필터·KPI·표           |
| REFRESH   | Real-time push (WebSocket) 실시간 푸시                |
| SNAPSHOT  | Daily 00:00 KST 매일 자정 스냅샷                        |
| TABLES    | wh_inventory , md_item , md_location , md_minmax |
| RBAC      | WH.View (read-only for most users)               |

02

Screen Mockup

화면 목업



## A-MES · Warehouse · Inventory Status

Warehouse &gt; Inventory

All Types

All Warehouses

All Status

Item Code / Name...

Search

Export Excel

Refresh

124

Total Items  
전체 품목 수

3

△ Below Min  
최소재고 미달

23

Low Stock  
재고 부족 주의

98

Normal  
정상

₩

42.7M

Total Value  
총 재고 금액

| Item Code<br>품목코드 | Item Name<br>품목명                      | Type<br>유형   | On Hand<br>현재고 | Unit | Min Stock<br>최소재고 | Δ vs Yesterday<br>전일 대비 | Location<br>위치 |
|-------------------|---------------------------------------|--------------|----------------|------|-------------------|-------------------------|----------------|
| RM-PP-NAT-20      | PP Pellet Natural 20MFI<br>PP 펠렛 나뉘럴  | Raw Material | 1,250 kg       | kg   | 300 kg            | ▲ 500 kg                | WH-RM-A-01-2   |
| FAB-GY-C03        | Fabric Roll Grey GY-C03<br>원단 롤 회색    | Fabric       | 42 roll        | roll | 15 roll           | ▲ 20 roll               | WH-FAB-D-01-1  |
| PNT-BK-NH578      | Black Paint NH578<br>블랙 도료 NH578      | Paint        | 284 L          | L    | 100 L             | ▼ 16 L                  | WH-PNT-E-01-1  |
| FAB-BK-C01        | Fabric Roll Black BK-C01<br>원단 롤 블랙   | Fabric       | 18 roll        | roll | 15 roll           | ▼ 4 roll                | WH-FAB-D-02-1  |
| RM-MB-GY-03       | Color Masterbatch Grey<br>착색 마스터배치 회색 | Raw Material | 0.5 kg         | kg   | 2 kg              | ▼ 0.3 kg                | WH-RM-A-02-1   |

**Row legend · 행 색상** — Green: Normal / Yellow: Low Stock (within 30% of min) / Red: Below minimum stock level. Δ column shows change since previous-day 00:00 snapshot.

## 03

## UI Component Specification

UI 컴포넌트 명세

## 3.1 Filter Toolbar · 필터 툴바

| ID     | COMPONENT                        | TYPE   | REQ | VALIDATION / BEHAVIOR                                                                           | DEFAULT        |
|--------|----------------------------------|--------|-----|-------------------------------------------------------------------------------------------------|----------------|
| CMP-01 | <b>Type Filter</b><br>유형 필터      | SELECT | N   | Options: All / Raw Material / Fabric / Paint / WIP . Multi-select disabled v1.                  | All Types      |
| CMP-02 | <b>Warehouse Filter</b><br>창고 필터 | SELECT | N   | Loaded from md_warehouse . WH-RM / WH-FAB / WH-PNT / WH-FG.                                     | All Warehouses |
| CMP-03 | <b>Status Filter</b><br>상태 필터    | SELECT | N   | Options: All / Below Min / Low Stock / Normal / Over Max . Below Min auto-applied on KPI click. | All Status     |
| CMP-04 | <b>Search Box</b><br>검색창         | TEXT   | N   | Min 2 chars. Search by item_code or item_name . Debounce 300ms.                                 | (empty)        |
| CMP-05 | <b>Refresh</b><br>새로고침           | BUTTON | N   | Force-fetch latest data ignoring WebSocket buffer.                                              | —              |
| CMP-06 | <b>Excel Export</b><br>엑셀 출력     | BUTTON | N   | Exports current filter result. Max 10,000 rows. Filename: WH03_Inventory_YYYYMMDD_HHmm.xlsx .   | —              |

### 3.2 KPI Strip · KPI

| ID     | METRIC                        | TYPE  | CALCULATION                                                                                    | DRILL-DOWN   |
|--------|-------------------------------|-------|------------------------------------------------------------------------------------------------|--------------|
| KPI-01 | <b>Total Items</b><br>전체 품목 수 | COUNT | COUNT(DISTINCT item_code) FROM wh_inventory WHERE on_hand_qty > 0 .                            | —            |
| KPI-02 | <b>Below Min</b><br>최소재고 미달   | COUNT | Count where on_hand_qty < md_minmax.min_qty . Color: #dc2626 . Click → filter status=BelowMin. | Filter apply |
| KPI-03 | <b>Low Stock</b><br>재고 부족     | COUNT | Count where on_hand_qty within 1.0× ~ 1.3× min_qty. Color: #f59e0b .                           | Filter apply |
| KPI-04 | <b>Normal</b><br>정상           | COUNT | Count where on_hand_qty ≥ 1.3 × min_qty . Color: #22c55e .                                     | Filter apply |
| KPI-05 | <b>Total Value</b><br>총 재고 금액 | SUM   | SUM(on_hand_qty × unit_cost) from md_item.unit_cost . Display in ₩.                            | —            |

### 3.3 Data Table Columns · 데이터 표 컬럼

| ID     | COLUMN                  | SOURCE                                | FORMAT / RULE                                                                         | SORT |
|--------|-------------------------|---------------------------------------|---------------------------------------------------------------------------------------|------|
| COL-01 | Item Code<br>품목코드       | <code>md_item.item_code</code>        | Monospace. Below-min items: bold red. Click → item master popup.                      | ASC  |
| COL-02 | Item Name<br>품목명        | <code>md_item.name_en/ko</code>       | EN main + KO subtext.                                                                 | —    |
| COL-03 | Type<br>유형              | <code>md_item.type</code>             | Badge: RM (gray) / Fabric (blue) / Paint (orange) / WIP (purple).                     | —    |
| COL-04 | On Hand<br>현재고          | <code>wh_inventory.on_hand_qty</code> | Right-align. Below min: red. Low: yellow. Normal: green. Decimal per unit.            | DESC |
| COL-05 | Unit<br>단위              | <code>md_item.unit</code>             | kg / roll / L / EA.                                                                   | —    |
| COL-06 | Min Stock<br>최소재고       | <code>md_minmax.min_qty</code>        | Right-align. Click → edit modal (Manager only).                                       | —    |
| COL-07 | Δ vs Yesterday<br>전일 대비 | <i>Calculated</i>                     | <code>on_hand - snapshot_prev_day</code> .<br>▲ green if positive, ▼ red if negative. | —    |
| COL-08 | Location<br>위치          | <code>wh_inventory.location_id</code> | If multiple locations, show count: <code>WH-FAB-D-01 (+2)</code> . Click → WH-04 map. | —    |
| COL-09 | Status<br>상태            | <i>Calculated</i>                     | Auto-derived from <code>on_hand</code> vs min/max thresholds.                         | —    |

## 04 Event & Action Flow

이벤트 · 액션 흐름

1

**USER** `page.Load` · WH-03 진입

User opens WH-03. Default filter (All types, all warehouses) applied. WebSocket subscribes to inventory updates.

WH-03 진입 시 기본 필터 적용 + WebSocket 재고 업데이트 구독.

**API** `GET /api/wh/inventory` + **WS** `/ws/inventory.subscribe`.

2

**SYSTEM** `push` · `inventory.updated`

When WH-02 inbound or WH-07 release commits, server pushes inventory delta to all subscribed clients. Row updates in place with brief highlight.

WH-02 입고·WH-07 출고 커밋 시 서버가 재고 변경분을 구독 클라이언트에 푸시. 해당 행이 잠시 하이라이트되며 갱신.

3

**USER** click · KPI [Below Min]

User clicks a KPI card. System auto-applies the corresponding filter and scrolls to the data table.

KPI 카드 클릭 시 해당 필터 자동 적용 + 데이터 표로 스크롤.

**UI** Status filter set, table re-rendered, deep-link URL: `?status=below_min`.

4

**SYSTEM** cron · 00:00 daily snapshot

Daily 00:00 KST job writes a full inventory snapshot to `wh_inventory_snapshot` for  $\Delta$  calculation and audit.

매일 00:00 KST 배치가 전체 재고 스냅샷을 `wh_inventory_snapshot`에 기록 (전일 대비 계산용).

**EFFECT**  $\Delta$  column populated next day; data retained 365 days.

5

**SYSTEM** trigger · min-stock alert

On every inventory commit, system checks if `on_hand` crossed below `min_qty` threshold. If so, generates alert + email to warehouse manager and procurement team.

재고 커밋 시마다 최소재고 임계 통과 여부 확인. 미달 시 창고관리자·구매팀에 알림 +이메일 발송.

**SIDE-EFFECT** INSERT `sys_alert`, dedupe within 24h to prevent spam.

✓

**USER** click · row → location

Clicking a Location cell navigates to WH-04 Location Map centered on the clicked location, with the item highlighted.

위치 셀 클릭 시 WH-04 로케이션 맵으로 이동(해당 위치 중심·품목 하이라이트).

**NAV** `/wh/location-map?focus=WH-FAB-D-01-1&item=FAB-GY-C03`.

## 05

## Exception & Error Handling

예외 · 에러 처리

**E-WH03-001**

### WebSocket Disconnected

WebSocket 단절

|          |                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------|
| CAUSE    | Browser idle > 5min, network issue, or server restart.<br>유희·네트워크·서버 재시작.                                     |
| DETECT   | No heartbeat received in 30 seconds.                                                                          |
| ACTION   | Yellow badge in toolbar: "Live updates paused." Auto-reconnect with exponential backoff (2s, 4s, 8s, 16s...). |
| RECOVERY | On reconnect, full reload triggered to sync any missed updates.                                               |

⚠ LIVE PAUSED — RECONNECTING

E-WH03-002

### Snapshot Missing

스냅샷 없음

|          |                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------|
| CAUSE    | Yesterday's snapshot job failed or DB cleanup removed data.<br>배치 실패·데이터 정리.                    |
| DETECT   | No row in <code>wh_inventory_snapshot</code> for <code>DATE = CURRENT_DATE - 1</code> .         |
| ACTION   | Δ column shows "—" with tooltip: "No snapshot available". KPIs and other columns work normally. |
| RECOVERY | SYS admin re-runs snapshot job manually from SYS-05 batch monitor.                              |

⚠ NO SNAPSHOT — Δ UNAVAILABLE

E-WH03-003

### Excel Export Overflow

엑셀 출력 초과

|          |                                                                                            |
|----------|--------------------------------------------------------------------------------------------|
| CAUSE    | Filtered result > 10,000 rows.<br>결과가 1만 행 초과.                                             |
| DETECT   | Server-side count check before file generation.                                            |
| ACTION   | Reject. Modal: "Result exceeds 10,000 rows. Narrow filter or use RPT-02 inventory report." |
| RECOVERY | Apply more specific filter or use Reports module.                                          |

⚠ TOO MANY ROWS — NARROW FILTER

E-WH03-004

### No Min/Max Defined

최소·최대 미설정

|          |                                                                                            |
|----------|--------------------------------------------------------------------------------------------|
| CAUSE    | Item has no row in <code>md_minmax</code> (master data incomplete).<br>기준정보 미완성.           |
| DETECT   | LEFT JOIN returns NULL for <code>min_qty</code> .                                          |
| ACTION   | Status column shows gray "—" badge with tooltip: "Min/Max not set. Configure in MD-09."    |
| RECOVERY | Master Data manager defines min/max in MD-09 → item appears in Below Min / Low Stock KPIs. |

i MIN/MAX NOT SET — SEE MD-09

E-WH03-005

### Negative Inventory

음수 재고

|          |                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------|
| CAUSE    | Race condition or data anomaly causes <code>on_hand_qty &lt; 0</code> .<br>동시성·데이터 이상.                           |
| DETECT   | Row-level check on load. Should never occur (BR-WH-05 enforces).                                                 |
| ACTION   | Highlight row with red border + warning icon. Auto-create incident ticket to SYS team. Send email to WH manager. |
| RECOVERY | WH-05 inventory adjustment with reason code <code>SYSTEM_ANOMALY</code> + supervisor approval.                   |

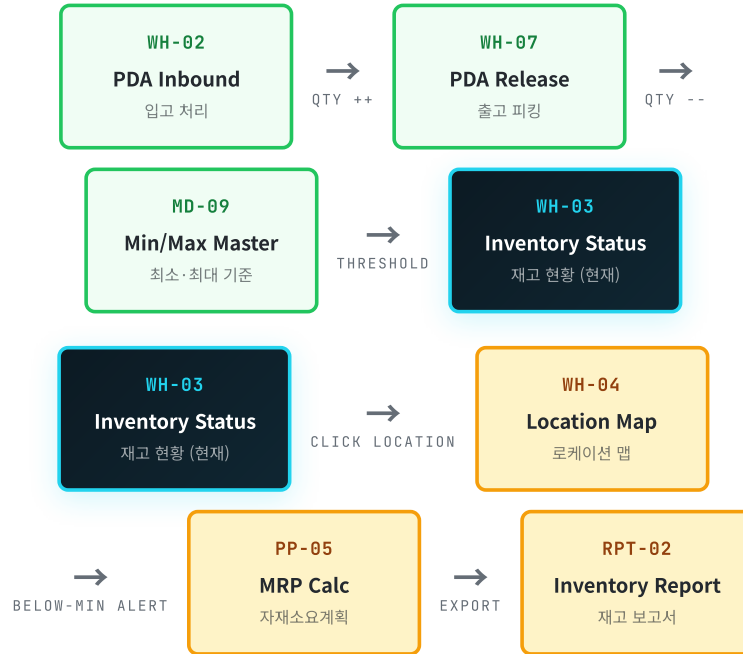
**E-WH03-006 Empty Result**  
조회 결과 없음

|          |                                                                               |
|----------|-------------------------------------------------------------------------------|
| CAUSE    | Filter combination yields zero items.<br>필터 조합으로 결과 없음.                       |
| DETECT   | API returns empty array.                                                      |
| ACTION   | Empty state illustration + "No items match your filter." Reset Filter button. |
| RECOVERY | User adjusts or resets filters.                                               |

i NO RESULTS — ADJUST FILTER

## 06 Linked Screens

선행 · 후행 연계 화면



## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                                                     | ACTION |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-03 | When on-hand qty falls below the minimum stock level (MD-09), an alert is sent to the warehouse manager and procurement team within 5 minutes.<br>현재고가 최소재고(MD-09) 미만이면 5분 이내 창고관리자·구매팀에 알림 발송. | Alert  |

| RULE ID  | DESCRIPTION                                                                                                                                                                            | ACTION |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-05 | Inventory quantity must never be negative. Any operation that would result in negative on-hand is rejected at DB constraint level.<br><br>재고 수량은 음수가 될 수 없으며, 음수 발생 작업은 DB 제약 조건으로 거부. | Block  |
| BR-WH-06 | Daily inventory snapshot at 00:00 KST is mandatory and retained for 365 days for audit and trend analysis.<br><br>매일 00:00 KST 재고 스냅샷 필수 + 감사·추이 분석용 365일 보관.                          | Auto   |

A-MESIndex

WH WarehousePP PlanINJ InjectionIMG WrappingPNT PaintingQC QualityFG Finished GoodsMNT Maintenance

L1A-MES Index > L2VOL.3 · WH Warehouse > L3WH-04 · Location Map

VOL.3 · WH MODULE

WAREHOUSE

창고 관리

L3 SCREENS · 화면 상세

WH-01Inbound Schedule입고 예정

WH-02PDA InboundPDA 입고 처리

WH-03Inventory Status재고 현황

WH-04Location Map로케이션 맵

WH-05Inventory Adjust재고 조정

WH-06Release Schedule출고 예정

WH-07PDA ReleasePDA 출고 피킹

WH-08Transaction History입출고 이력

ON THIS PAGE

01Overview

02Screen Mockup

03UI Components

04Event & Action Flow

05Exception Handling

06Linked Screens

07Business Rules

VOL.3 · WH · WAREHOUSE MANAGEMENT · 로케이션 맵

WH-04

LOCATION MAP

로케이션 맵

DEVICEWeb PC

PHASE2nd Dev

ROLEWarehouse Staff · Manager

FREQOn-demand · 필요시

URL/wh/location-map

01Overview

화면 개요

Visual grid map of all warehouse zones and shelf locations. Each cell represents one storage cell and is color-coded by fill rate: empty (grey), partial (yellow), full (green), or over-capacity (red). Clicking a cell opens a detail panel with stored items, quantities, inbound dates, and a link to the cell's stock movement history. Location codes follow the convention [Warehouse] - [Zone] - [Row] - [Col] - [Level] .

모든 창고 구역과 선반 로케이션을 그리드 형태로 시각화한 화면입니다. 각 셀은 한 개의 저장 위치를 나타내며 충전율에 따라 색상이 다릅니다: 비어있음(회색), 일부 사용(노랑), 가득 참(초록), 초과(빨강). 셀 클릭 시 보관 품목·수량·입고일·이력 링크가 포함된 상세 패널이 열립니다. 로케이션 코드 체계: [창고]-[구역]-[행]-[열]-[층] .

|             |                                                |
|-------------|------------------------------------------------|
| SCREEN ID   | WH-04                                          |
| LAYOUT      | Multi-zone visual grid + side panel  다중 구역 그리드 |
| REFRESH     | Real-time (WebSocket)  실시간                     |
| PHASE       | 2nd Dev  2차 개발                                 |
| CODE FORMAT | WH-FAB-D-01-1                                  |
| TABLES      | md_location , wh_inventory                     |



## 02 Screen Mockup

화면 목업 — 시각적 그리드 맵

● WH-04 · LOCATION MAP · WEB PC

A-MES · Warehouse · Location Map

Warehouse &gt; Location Map

All Warehouses ▾

All Levels · 전체 층 ▾

Location code search...

Search

Print QR Labels

Excel

240

Total Cells  
전체 셀

38

Empty  
비어있음

52

Partial  
일부 사용

148

Full  
가득 참

2

Over Cap  
초과

Warehouse Map

View · 창고 맵 (Level 1)

Empty  
· 비어있  
음Partial  
· 일부Full  
· 가  
득Over  
· 초과

▼ WH-RM-A · Raw Material Zone A · 원자재 A구역

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| A-01-1 | A-02-1 | A-03-1 | A-04-1 | A-05-1 | A-06-1 | A-07-1 | A-08-1 | A-09-1 | A-10-1 |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

▼ WH-FAB-D · Fabric Zone D · 원단 D구역 (2 rows × 10 cols)

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| D-01-1 | D-02-1 | D-03-1 | D-04-1 | D-05-1 | D-06-1 | D-07-1 | D-08-1 | D-09-1 | D-10-1 |
| D-01-2 | D-02-2 | D-03-2 | D-04-2 | D-05-2 | D-06-2 | D-07-2 | D-08-2 | D-09-2 | D-10-2 |

▼ WH-PNT-E · Paint Zone E · 도료 E구역

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| E-01-1 | E-02-1 | E-03-1 | E-04-1 | E-05-1 | E-06-1 | E-07-1 | E-08-1 | E-09-1 | E-10-1 |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

Selected Cell · 선택 셀:

WH-FAB-D-01-1

Zone **Fabric Zone D · Row 1 / Col 1 / Level 1**

Capacity **50 rolls**

On Hand **42 rolls (84%)**

Item **FAB-GY-C03**

FIFO  
Date **2025-07-01 09:14**

Recent Activity · 최근 거래

▲ **IN** · 20 rolls · 2025-07-01  
09:14 · 김기사

▼ **OUT** · 5 rolls · 2025-07-02  
14:22 · 박작업

▲ **IN** · 30 rolls · 2025-07-03  
11:05 · 김기사

[View full history → WH-08](#)

**i** **Fill rate · 충전율 계산** — empty: 0% · partial: 1–80% · full: 80–100% · over: >100% (capacity exceeded). Hover any cell for tooltip with item code and qty.

## 03 UI Component Specification

UI 컴포넌트 명세

### 3.1 Filter Toolbar · 필터

| ID     | COMPONENT                          | TYPE   | REQ | VALIDATION / BEHAVIOR                                                                                               | DEFAULT |
|--------|------------------------------------|--------|-----|---------------------------------------------------------------------------------------------------------------------|---------|
| CMP-01 | <b>Warehouse Filter</b><br>창고 필터   | SELECT | N   | Loaded from <code>md_warehouse</code> .<br>Selecting a warehouse shows only its zones.                              | All     |
| CMP-02 | <b>Level Filter</b><br>층 필터        | SELECT | N   | For multi-level racks.<br>Filters by <code>md_location.level</code> .                                               | L1      |
| CMP-03 | <b>Location Search</b><br>로케이션 검색  | TEXT   | N   | Searches by <code>location_code</code> or stored <code>item_code</code> .<br>Match → cell flashes + zoom-in.        | (empty) |
| CMP-04 | <b>Print QR Labels</b><br>QR 라벨 출력 | BUTTON | N   | Opens print modal.<br>Select cells (or zone) → batch-print location QR labels.<br>Requires <code>WH.Manage</code> . | —       |
| CMP-05 | <b>Excel Export</b><br>엑셀 출력       | BUTTON | N   | Exports cell inventory list. Includes capacity, on-hand, fill rate %.                                               | —       |

3.2 Grid Cell Component · 그리드 셀

| ID     | COMPONENT                   | TYPE    | REQ | VALIDATION / BEHAVIOR                                                                                         | DEFAULT |
|--------|-----------------------------|---------|-----|---------------------------------------------------------------------------------------------------------------|---------|
| CMP-06 | Cell Color State<br>셀 색상 상태 | DERIVED | -   | Auto: 0% → grey · 1-80% → yellow · 80-100% → green · >100% → red.                                             | -       |
| CMP-07 | Cell Hover Tooltip<br>호버 툴팁 | TOOLTIP | -   | Shows: location_code, item_code, on_hand / capacity, fill %, FIFO date. 200ms delay.                          | -       |
| CMP-08 | Cell Click<br>셀 클릭          | CLICK   | -   | Opens side panel with full details + activity log. URL updates with <code>?focus=...</code> for deep linking. | -       |
| CMP-09 | Empty Cell Action<br>빈 셀 클릭 | CLICK   | -   | For empty cells, panel shows "Available" state + capacity. Manager can flag as "blocked/maintenance".         | -       |

3.3 Detail Side Panel · 상세 패널

| ID     | FIELD                         | SOURCE                       | FORMAT                                            | EDITABLE       |
|--------|-------------------------------|------------------------------|---------------------------------------------------|----------------|
| FLD-01 | Location Code<br>위치 코드        | md_location.code             | Monospace, cyan badge.                            | No             |
| FLD-02 | Zone / Row / Col / Level      | md_location                  | Decomposed display.                               | No             |
| FLD-03 | Capacity<br>용량                | md_location.capacity         | Number + unit (rolls/kg/L/EA per item type).      | Manager        |
| FLD-04 | On Hand / Fill %<br>현재고 · 충전율 | wh_inventory                 | Number + percentage. Color matches cell state.    | No (use WH-05) |
| FLD-05 | FIFO Date<br>FIFO 기준 일        | MIN(wh_inbound.inbound_date) | Oldest inbound date in the cell. Datetime format. | No             |

| ID     | FIELD                           | SOURCE             | FORMAT                                                                     | EDITABLE |
|--------|---------------------------------|--------------------|----------------------------------------------------------------------------|----------|
| FLD-06 | <b>Recent Activity</b><br>최근 거래 | wh_transaction     | Last 5 entries:<br>IN/OUT, qty, time, operator.<br>Link to WH-08 for full. | No       |
| FLD-07 | <b>Location Status</b><br>위치 상태 | md_location.status | Active / Blocked / Maintenance.<br>Manager can toggle.                     | Manager  |

## 04 Event & Action Flow

이벤트 · 액션 흐름

1

**USER** page.load · WH-04 진입

User opens WH-04. System loads location master + current inventory per location. Renders grid color-coded by fill rate. WebSocket subscribes to inventory deltas.

WH-04 진입 시 위치 마스터 + 현재 재고를 로딩하여 충전율 색상 그리드 렌더링. WebSocket 구독.

```
API GET /api/wh/locations · WS /ws/location.subscribe.
```

2

**USER** hover · cell

User hovers a cell. Tooltip appears after 200ms with item code, qty, fill %, FIFO date.

셀 호버 시 200ms 지연 후 툴팁 표시(품목·수량·충전율·FIFO 일자).

**UI** No network call · uses cached data from initial load.

3

**USER** click · cell

User clicks a cell. Side panel opens with full detail + recent activity. URL updates with ?focus=WH-FAB-D-01-1 for shareable deep link.

셀 클릭 시 상세 패널이 열리고 최근 거래 표시. URL이 ?focus=...로 갱신되어 딥링크 공유 가능.

```
API GET /api/wh/location/:code/detail.
```

4

**SYSTEM** push · location.updated

When WH-02 inbound or WH-07 release commits, server pushes the location delta. Cell color and tooltip update with smooth transition animation.

WH-02-WH-07 커밋 시 위치 변경분 푸시. 셀 색상·툴팁이 부드러운 전환과 함께 갱신.

```
PAYLOAD {location_code, on_hand, capacity, fill_rate}.
```

5

USER click · [Print QR Labels]

Manager selects one or more cells (or entire zone) and clicks Print. System generates a printable HTML page with QR codes for each location.

매니저가 셀(또는 구역)을 선택하고 [QR 출력] 클릭 시 QR 코드 포함 인쇄 페이지 생성.

```
REQUIRE Role WH.Manage · QR encodes location_code · ZPL or HTML format.
```

✓

USER click · "View history"

From side panel, user clicks "View full history" link.

Navigates to WH-08 with location pre-filter applied.

상세 패널에서 [전체 이력] 클릭 시 WH-08로 이동(해당 위치 필터 사전 적용).

```
NAV /wh/history?location=WH-FAB-D-01-1.
```

## 05

### Exception & Error Handling

예외 · 에러 처리

E-WH04-001

#### Over-Capacity Detected

초과 적재 감지

|          |                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------|
| CAUSE    | Cell's on_hand exceeds defined capacity (data anomaly or manual override).<br>정의 용량 초과(이상·수동 입력). |
| DETECT   | Cell render: <code>on_hand &gt; capacity</code> .                                                 |
| ACTION   | Cell color = red. KPI "Over Cap" increments. Alert to WH manager. Recommend relocation.           |
| RECOVERY | Manager redistributes via WH-05 or expands capacity in MD-11.                                     |

OVER CAPACITY — RELOCATE

E-WH04-002

#### Location Blocked

## 위치 차단

|          |                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------|
| CAUSE    | Cell marked as <code>Blocked</code> or <code>Maintenance</code> in MD-11.<br>차단·정비 상태.                |
| DETECT   | <code>md_location.status ≠ 'Active'</code> .                                                          |
| ACTION   | Cell shown with diagonal stripe pattern. WH-02/WH-07 reject scanning this location until reactivated. |
| RECOVERY | Manager re-activates location in MD-11.                                                               |

⚠ BLOCKED — SEE MD-11

E-WH04-003

## Mixed Items in Cell

셀 내 품목 혼재

|          |                                                                                               |
|----------|-----------------------------------------------------------------------------------------------|
| CAUSE    | Multiple item codes stored in same cell (violates BR-WH-07 if not allowed).<br>한 셀에 복수 품목 보관. |
| DETECT   | <code>COUNT(DISTINCT item_code) &gt; 1</code> per location.                                   |
| ACTION   | Cell badge shows "Mixed". Detail panel lists all items separately. Yellow warning icon.       |
| RECOVERY | If policy disallows, WH-05 adjustment to relocate.                                            |

⚠ MIXED ITEMS — REVIEW

E-WH04-004

## QR Print Permission Denied

QR 출력 권한 없음

|          |                                                                                       |
|----------|---------------------------------------------------------------------------------------|
| CAUSE    | Non-manager attempts to print QR labels.<br>매니저 권한 없는 사용자가 출력 시도.                     |
| DETECT   | Server RBAC check on print request.                                                   |
| ACTION   | Return 403. Modal: "QR label printing requires WH.Manage role. Contact your manager." |
| RECOVERY | Request from manager or escalate.                                                     |

🚫 ACCESS DENIED — WH.MANAGE REQUIRED

E-WH04-005

## Empty Warehouse View

빈 창고 뷰

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| CAUSE    | Selected warehouse has no defined locations in MD-11.<br>선택 창고에 위치 미정의.           |
| DETECT   | Locations query returns empty.                                                    |
| ACTION   | Show illustration: "No locations defined for this warehouse. Configure in MD-11." |
| RECOVERY | Master data manager defines locations.                                            |

E-WH04-006

## Rendering Performance

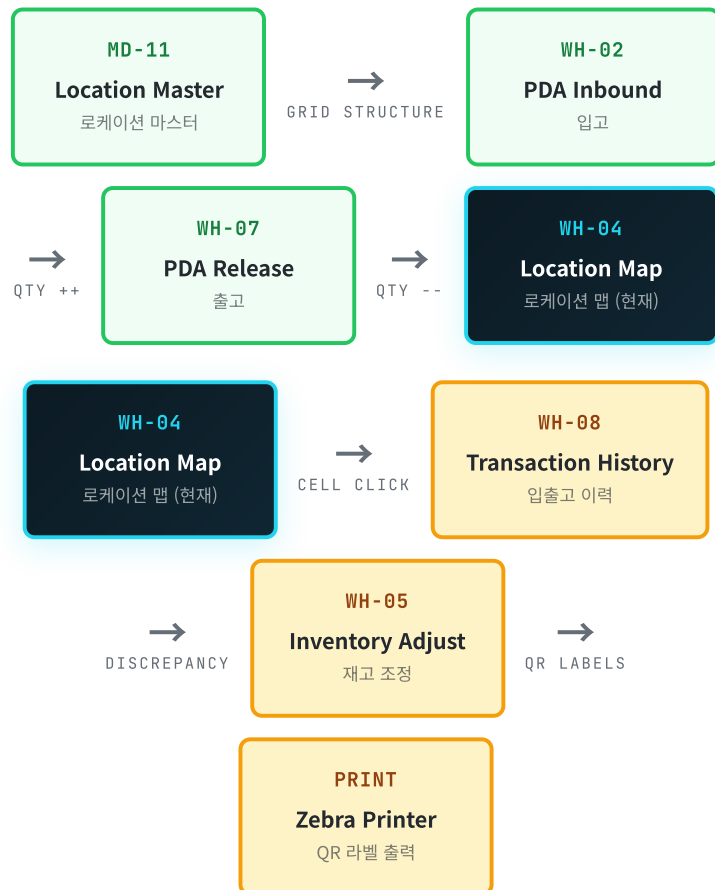
렌더링 성능

|          |                                                                                               |
|----------|-----------------------------------------------------------------------------------------------|
| CAUSE    | Large warehouse (> 1,000 cells) causes slow render.<br>대형 창고(1000셀 초과) 렌더링 지연.                |
| DETECT   | Cell count check at load time.                                                                |
| ACTION   | Activate virtualized rendering (only visible cells). Show progress bar during initial render. |
| RECOVERY | User applies warehouse/level filter to reduce scope.                                          |

! LARGE GRID - VIRTUALIZED

## 06 Linked Screens

선행 · 후행 연계 화면



## 07 Applicable Business Rules

| RULE ID  | DESCRIPTION                                                                                                                                                                                                             | ACTION |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-07 | <p>Each storage cell should hold a single item code. Mixed-item cells require manager approval and are flagged in the map view.</p> <p>각 셀은 단일 품목 보관 원칙. 혼재 시 매니저 승인 + 맵에 플래그 표시.</p>                                   | Warn   |
| BR-WH-08 | <p>Blocked or maintenance-status locations cannot receive new inbound. WH-02 / WH-05 scans are rejected until reactivated.</p> <p>차단·정비 상태 위치 신규 입고 불가. WH-02·WH-05 스캔 거부.</p>                                          | Block  |
| BR-WH-09 | <p>Location capacity is defined in MD-11 and is hard-enforced at WH-02 receiving time. Over-capacity stocking is rejected unless override approved.</p> <p>로케이션 용량은 MD-11에 정의되며 WH-02 입고 시 강제 적용. 초과 시 거부(승인 시 예외).</p> | Block  |



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PHASE

1st Dev

ROLE

WH.Adjust + WH.Approve

FREQ

As needed · 수시

URL

/wh/adjustment

01

Overview

화면 개요

Manual correction of system inventory to match physical count results. Every adjustment requires a **reason code** (one of: `COUNT_DIFF` , `DAMAGE` , `EXPIRED` , `SYSTEM_ANOMALY` , `RELOCATION` ) and **supervisor approval** before commit. Adjustments above ₩500,000 in value or 10% of on-hand qty automatically escalate to plant manager approval. All adjustments are written to WH-08 transaction history with before/after values, photos (optional), and approver signature.

실사 결과를 시스템 재고에 반영하는 수동 조정 화면입니다. 모든 조정은 사유 코드 ( `COUNT_DIFF` 실사 차이 · `DAMAGE` 파손 · `EXPIRED` 유통기한 · `SYSTEM_ANOMALY` 시스템 이상 · `RELOCATION` 위치 이동)와 관리자 승인 후 확정됩니다. 50만원 초과 또는 현재고 10% 초과 조정은 공장장 승인으로 자동 에스컬레이션. 모든 조정은 전/후 값, 사진(선택), 승인자 서명과 함께 WH-08에 기록됩니다.

|           |                                                       |
|-----------|-------------------------------------------------------|
| SCREEN ID | WH-05                                                 |
| LAYOUT    | Search + Adjustment form + Approval queue 검색·조정폼·승인대기 |
| APPROVAL  | 2-level (Supervisor + Plant Mgr) 2단계 승인               |
| THRESHOLD | ₩500K or 10% qty → escalate 자동 에스컬레이션                 |
| TABLES    | wh_adjustment , wh_inventory , wh_transaction         |
| RBAC      | WH.Adjust · approver: WH.Approve                      |
| PHASE     | 1st Dev                                               |

● WH-05 · INVENTORY ADJUSTMENT · WEB PC

A-MES · Warehouse · Inventory Adjustment

Warehouse &gt; Adjustment

+ New Adjustment · 신규 조정

All Status

Item / Location / Adj.ID...

Search

Excel

4

Pending Approval  
승인 대기

1

Escalated  
에스컬레이션

38

Approved · 30d  
30일 승인

₩

2.4M

Adj. Value · 30d  
30일 조정금액NEW ADJUSTMENT  
FORM · 신규 조정 입력Item Code  
\* FAB-GY-C03

Location \* WH-FAB-D-01-1

Current Qty  
42 rollsCounted Qty \*  
40 rollsΔ Adjustment  
-2 rolls (-4.8%)Value Impact  
-₩ 86,000Reason Code \*  
COUNT\_DIFFComment  
2025-Q3 실사  
결과 2 roll 부

Attach Photo

Photo (선택) DAMAGE/EXPIRED 시  
필수Save  
DraftSubmit for  
Approval

## PENDING APPROVAL · 승인 대기 목록

ADJ-2025-0715-001

Pending · 대기

FAB-GY-C03 · WH-FAB-D-01-1 · -2  
rolls (-₩86,000)Reason: COUNT\_DIFF · Requested  
by: 김기사 · 15:32✓  
Approve✗  
RejectView  
Detail

ADJ-2025-0715-002

Escalated · 에스컬레이션

RM-PP-NAT-20 · WH-RM-A-01-2 ·  
-45 kg (-₩630,000)Reason: DAMAGE · 지게차 충돌 분진  
발생 · Plant Mgr approval requiredWaiting Plant  
MgrView  
Detail

ADJ-2025-0715-003 Draft · 작성중

PNT-BK-NH578 · WH-PNT-E-01-1 ·  
+3 LReason: RELOCATION · Draft by: 박  
작업

**Auto-escalation rules · 자동 에스컬레이션** — 조정 절대값이 50만원 초과 또는 현재고의 10% 초과인 경우 자동으로 공장장(Plant Manager) 승인으로 전달됩니다. Supervisor만으로는 승인 불가.

## 03

## UI Component Specification

UI 컴포넌트 명세

## 3.1 Adjustment Form Fields · 조정 품 입력 필드

| ID     | COMPONENT                    | TYPE         | REQ | VALIDATION / BEHAVIOR                                                                                                                                                                 | DEFAULT      |
|--------|------------------------------|--------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| CMP-01 | Item Code<br>품목코드            | AUTOCOMPLETE | Y   | Search from <code>md_item</code> . Min 2 chars. Selecting auto-loads location list with on-hand qty.                                                                                  | (empty)      |
| CMP-02 | Location<br>위치               | SELECT       | Y   | Locations where item has on-hand. If empty cell, requires item type compatibility check.                                                                                              | (empty)      |
| CMP-03 | Current Qty<br>현재 수량         | DISPLAY      | —   | Read-only. From <code>wh_inventory.on_hand_qty</code> . Updates live if changed by other transactions.                                                                                | —            |
| CMP-04 | Counted Qty<br>실사 수량         | NUMBER       | Y   | Must be $\geq 0$ . Decimal per item unit. Cannot equal current qty (otherwise no adjustment needed).                                                                                  | (empty)      |
| CMP-05 | $\Delta$ Adjustment<br>조정량   | CALCULATED   | —   | <code>counted - current</code> . Display sign + percentage. Red if negative, green if positive.                                                                                       | —            |
| CMP-06 | Value Impact<br>금액 영향        | CALCULATED   | —   | <code><math>\Delta \times md\_item.unit\_cost</math></code> . KRW format. Triggers escalation check on input change.                                                                  | —            |
| CMP-07 | Reason Code<br>사유 코드         | SELECT       | Y   | Options: <code>COUNT_DIFF</code> / <code>DAMAGE</code> / <code>EXPIRED</code> / <code>SYSTEM_ANOMALY</code> / <code>RELOCATION</code> . <code>DAMAGE/EXPIRED</code> → photo required. | (unselected) |
| CMP-08 | Comment<br>상세 사유             | TEXTAREA     | Y   | Min 10 chars. Max 500. Free-text describing the reason in detail.                                                                                                                     | (empty)      |
| CMP-09 | Photo Attachment<br>사진 첨부    | FILE         | N*  | JPEG/PNG, max 5MB, up to 3 files. Required when reason = <code>DAMAGE</code> or <code>EXPIRED</code> .                                                                                | (none)       |
| CMP-10 | Save Draft<br>임시 저장          | BUTTON       | N   | Saves with status=Draft. Does not commit inventory. Can be edited/resumed later.                                                                                                      | —            |
| CMP-11 | Submit for Approval<br>승인 요청 | BUTTON       | Y   | Validates all required fields. Status → Pending. Routes to supervisor (or plant mgr if escalated).                                                                                    | —            |

## 3.2 Approval Queue Actions · 승인 대기 액션

| ID     | COMPONENT                    | TYPE   | ROLE | VALIDATION / BEHAVIOR                                                                             | RESULT    |
|--------|------------------------------|--------|------|---------------------------------------------------------------------------------------------------|-----------|
| CMP-12 | [✓ Approve]<br>승인            | BUTTON | SUP  | Only enabled if not escalated. Requires confirmation dialog. Server commits inventory atomically. | Committed |
| CMP-13 | [X Reject]<br>반려             | BUTTON | SUP  | Requires rejection reason. Status → Rejected. Notify requester via in-app + email.                | Rejected  |
| CMP-14 | [View Detail]<br>상세 보기       | BUTTON | ALL  | Opens read-only modal with full form data, photos, history.                                       | —         |
| CMP-15 | Plant Mgr Approval<br>공장장 승인 | BUTTON | PM   | Only for escalated adjustments. Plant Mgr signs (PIN or biometric on tablet). Audit logged.       | Committed |

### 3.3 Reason Code Definitions · 사유 코드 정의

| CODE       | NAME                      | DEFINITION                                                                                   | PHOTO REQ. | ESCALATION      |
|------------|---------------------------|----------------------------------------------------------------------------------------------|------------|-----------------|
| COUNT_DIFF | Count Difference<br>실사 차이 | Discrepancy between physical count and system qty without obvious cause. Most common reason. | No         | By threshold    |
| DAMAGE     | Damaged Stock<br>파손       | Physical damage from handling, dropping, contamination. Photo evidence mandatory.            | Yes        | Always escalate |
| EXPIRED    | Expired Stock<br>유통기한 경과  | Materials past shelf life (e.g. paint,                                                       | Yes        | By threshold    |

| CODE           | NAME                            | DEFINITION                                                                                   | PHOTO REQ. | ESCALATION      |
|----------------|---------------------------------|----------------------------------------------------------------------------------------------|------------|-----------------|
|                |                                 | adhesives).<br>Photo + batch info required.                                                  |            |                 |
| SYSTEM_ANOMALY | <b>System Anomaly</b><br>시스템 이상 | Data corruption, race condition, or known bug correction. Requires SYS team incident ticket. | No         | Always escalate |
| RELOCATION     | <b>Location Move</b><br>위치 이동   | Internal transfer between locations (paired entry: + and - in two locations). Same value.    | No         | No              |

## 04 Event & Action Flow

이벤트 · 액션 흐름

1

**USER** click · [+ New Adjustment]

Warehouse staff opens new adjustment form. Requires WH.Adjust role.

창고 담당자가 신규 조정 품을 엽니다. WH.Adjust 권한 필요.

**UI** Empty form rendered · item code autocomplete focused.

2

**USER** select · item + location

User selects item and location. System fetches current qty and unit cost. Form recalculates  $\Delta$  and value impact live as counted qty is entered.

품목·위치 선택 시 현재고와 단가 자동 로드. 실사 수량 입력에 따라  $\Delta$ ·금액 실시간 계산.

**API** GET /api/wh/inventory?item=...&loc=... .

3

**USER** select · reason code

User selects reason. If DAMAGE/EXPIRED, photo upload becomes required. If SYSTEM\_ANOMALY, incident ticket ID field appears.

사유 선택 시 DAMAGE/EXPIRED는 사진 필수, SYSTEM\_ANOMALY는 인시던트 티켓 ID 입력 필드 표시.

**UI** Form fields conditionally enabled/required.

4

**USER** click · [Submit for Approval]

All validations pass. System checks escalation thresholds: if  $|value| > ₩500K$  or  $|\Delta/current| > 10\%$ , routes to Plant Manager; else to

Supervisor.

전체 검증 통과 후 에스컬레이션 임계 확인: 금액 50만원 초과 또는 비율 10% 초과  
→ 공장장, 그 외 → 슈퍼바이저로 라우팅.

```
API POST /api/wh/adjustment · status='Pending' ·  
approver_level=1|2
```

5

SYSTEM notify · approver

Approver receives in-app notification + email. Approval queue card appears on their WH-05 view. SLA: 4 business hours for Supervisor, 1 business day for Plant Mgr.

승인자에게 인앱·이메일 알림 + 승인 대기 카드 표시. SLA: 슈퍼바이저 4시간, 공장장 1영업일.

SIDE-EFFECT SLA timer starts · auto-reminder at 75% elapsed.

6

APPROVER click · [✓ Approve]

Approver reviews detail + photos. Clicks Approve. System asks confirmation, then commits inventory change atomically in a transaction.

승인자가 상세·사진 확인 후 [승인] 클릭. 확인 다이얼로그 후 트랜잭션으로 재고 변경 원자적 커밋.

```
API POST /api/wh/adjustment/:id/approve · BEGIN TRAN;  
UPDATE wh_inventory; INSERT wh_transaction; COMMIT;
```

✓

SYSTEM complete · downstream

Inventory updated. WH-08 transaction logged with reason code, approver signature, before/after values. Requester notified. Below-min check re-evaluated.

재고 갱신. WH-08에 사유·승인자 서명·전후 값 기록. 요청자에게 결과 알림. 최소재고 재평가.

DOWNSTREAM WH-03/WH-04 push update · WH-08 audit row · BR-WH-03 re-check.

## 05

### Exception & Error Handling

예외 · 에러 처리

**E-WH05-001** Counted Qty = Current  
실사·현재 동일

CAUSE User enters counted qty equal to current. No adjustment needed.  
실사 수량이 현재고와 동일.

DETECT Form validation on Submit.

ACTION Block submit. Modal: "No adjustment needed — counted qty equals current. Cancel or correct?"

RECOVERY User cancels or enters different qty.

⊖ NO Δ — CANCEL OR CORRECT

E-WH05-002

### Negative Result Stock

음수 재고 결과

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| CAUSE    | Counted qty < 0, or system on_hand would become negative on commit.<br>결과 재고가 음수. |
| DETECT   | Server validation; DB CHECK constraint also enforces (BR-WH-05).                  |
| ACTION   | Block. "Counted qty must be $\geq$ 0. Inventory cannot be negative."              |
| RECOVERY | User re-counts physical stock.                                                    |

⊖ NEGATIVE NOT ALLOWED

E-WH05-003

### Photo Required (Missing)

사진 누락

|          |                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------|
| CAUSE    | Reason is DAMAGE or EXPIRED but no photo attached.<br>파손·만료 사유인데 사진 미첨부.                             |
| DETECT   | Form validation on Submit.                                                                           |
| ACTION   | Block submit. Highlight photo field red. Inline error: "Photo evidence required for DAMAGE/EXPIRED." |
| RECOVERY | User attaches photo or changes reason.                                                               |

⚠ PHOTO REQUIRED

E-WH05-004

### Concurrent Modification

동시 수정 충돌

|          |                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------|
| CAUSE    | Inventory changed (e.g., WH-02 commit) between form open and submit.<br>폼 열림과 제출 사이에 재고 변경.          |
| DETECT   | Optimistic lock: <code>version</code> column mismatch.                                               |
| ACTION   | Reject submit. Modal: "Inventory changed. Current qty is now X. Re-count or adjust?" Show new value. |
| RECOVERY | User refreshes data and reconsiders.                                                                 |

⚠ STALE DATA — REFRESH

E-WH05-005

### Approval SLA Breached

승인 SLA 초과

|          |                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------|
| CAUSE    | Approval pending past SLA (4hr Supervisor / 1day Plant Mgr).<br>SLA 시간 초과.                              |
| DETECT   | Hourly cron checks <code>pending_since</code> .                                                         |
| ACTION   | Auto-escalate to next level (Supervisor → Plant Mgr, Plant Mgr → Plant Director).<br>Email all parties. |
| RECOVERY | Next-level approver acts on request.                                                                    |

E-WH05-006

### Self-Approval Attempt

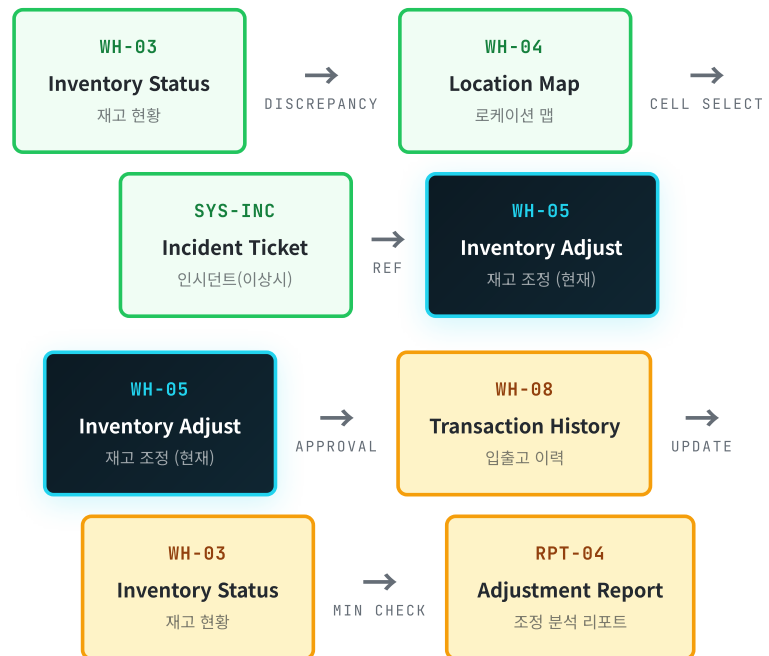
자체 승인 시도

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| CAUSE    | Approver attempts to approve own adjustment request.<br>요청자가 자신의 요청을 승인 시도.     |
| DETECT   | Server check: <code>approver_id == requester_id</code> .                        |
| ACTION   | Block. "Self-approval is not allowed. Route to another approver." Audit logged. |
| RECOVERY | Different approver handles request.                                             |

— SELF-APPROVAL FORBIDDEN

## 06 Linked Screens

선행 · 후행 연계 화면



## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                     | ACTION |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-05 | Inventory quantity must never be negative. DB-level constraint rejects any adjustment that would result in negative on-hand.<br>재고는 음수 불가. DB 제약으로 음수 결과 조정 거부. | Block  |



| RULE ID  | DESCRIPTION                                                                                                                                                                                                                         | ACTION   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| BR-WH-10 | All inventory adjustments require reason code + supervisor approval. Adjustments with  value  > ₩500K or > 10% of on-hand automatically escalate to plant manager.<br>모든 재고 조정은 사유코드 + 슈퍼바이저 승인 필수. 50만원 또는 10% 초과 시 공장장 자동 에스컬레이션. | Workflow |
| BR-WH-11 | DAMAGE and EXPIRED reasons require photo evidence (min 1, max 3 photos). Photos retained 5 years for audit.<br>파손·만료 사유는 사진 증빙 필수(1~3장). 감사용 5년 보관.                                                                                 | Required |
| BR-WH-12 | Self-approval is strictly forbidden. The requester cannot be the approver for the same adjustment.<br>자체 승인 금지. 요청자와 승인자가 동일인 불가.                                                                                                   | Block    |

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WAREHOUSE

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VOL.3 · WH · WAREHOUSE MANAGEMENT · 출고 예정

WH-06

RELEASE SCHEDULE

출고 예정

DEVICEWeb PC

PHASE1st Dev

ROLEWarehouse Staff (WH.Release)

FREQContinuous · 상시

URL/wh/release/schedule

01

Overview

화면 개요

Lists upcoming material release demands derived from active Work Orders (WO) in the PP module. For each WO line, the system computes BOM-exploded material needs vs current on-hand and flags shortages. Warehouse staff use this screen to **pre-pick** materials ahead of production start (preparation status), and click **[Start Picking]** to launch WH-07 PDA flow. Filters by line, shift, WO number, and item. ETA is the planned WO start time minus a configurable lead (default: 30 min).

PP 모듈 활성 작업지시(WO) 기반 자재 출고 예정 목록입니다. WO별로 BOM 분해된 자재 소요량을 현재고와 대조하여 부족 자재를 표시합니다. 창고 담당자는 본 화면에서 생산 시작 전 **사전 피킹**을 준비하고(준비 상태), **[Start Picking]** 클릭 시 WH-07 PDA 흐름이 시작됩니다. 라인·교대·WO 번호·품목 필터 지원. ETA는 WO 계획 시작 시각에서 설정된 리드타임(기본 30분)을 차감한 값입니다.

|           |                                          |
|-----------|------------------------------------------|
| SCREEN ID | WH-06                                    |
| LAYOUT    | Filter + KPI + Demand table   필터·KPI·수요표 |
| SOURCE    | PP-04 WO + PP-BOM   작업지시·BOM             |
| REFRESH   | Real-time push + 5 min cron   실시간 + 5분   |
| LEAD TIME | 30 min (config in MD-15)   기본 30분        |
| TABLES    | wo_header , wo_bom_demand , wh_inventory |

## 02

## Screen Mockup

화면 목업

● WH-06 · RELEASE SCHEDULE · WEB PC · 1920×1080

A-MES · Warehouse · Release  
ScheduleWarehouse > Release >  
Schedule

All Lines

All Shifts

All Status

연도-월-일

WO No / Item...

Search

Excel

23

Today's  
WOs  
금일 WO

14

Ready to  
Pick  
피킹 가능

6

Within 30  
min  
30분 이내 시  
작

2

△  
Shortage  
자재 부족

1

Released  
출고 완료

| WO<br>No.<br>작업<br>지시        | Line<br>·<br>Shift<br>라인·<br>교대 | Finished<br>Item<br>생산 품목             | WO<br>Qty | Material<br>to<br>Release<br>출고 자재 | Required<br>소요량 | On<br>Hand<br>현재고 | Pla<br>Sta<br>계획<br>작 |
|------------------------------|---------------------------------|---------------------------------------|-----------|------------------------------------|-----------------|-------------------|-----------------------|
| WO-<br>2025-<br>0715-<br>001 | INJ-<br>L1 ·<br>주간              | TRIM-<br>LH-001<br>트림 LH              | 2,030     | RM-PP-<br>NAT-20                   | 81 kg           | 1,250<br>kg       | 14                    |
| WO-<br>2025-<br>0715-<br>002 | IMG-<br>L1 ·<br>주간              | TRIM-<br>LH-001-<br>GY<br>트림 LH<br>회색 | 2,030     | FAB-GY-<br>C03                     | 12 roll         | 42<br>roll        | 16                    |
| WO-<br>2025-<br>0715-<br>003 | PNT-<br>L1 ·<br>주간              | CONS-<br>UP-001<br>콘솔 어퍼              | 650       | PNT-BK-<br>NH578                   | 320 L           | 284 L<br>△        | 18                    |
| WO-<br>2025-<br>0715-<br>004 | INJ-<br>L2 ·<br>야간              | TRIM-<br>RH-001<br>트림 RH              | 2,030     | RM-PP-<br>NAT-20                   | 81 kg           | 1,169<br>kg*      | 22                    |
| WO-<br>2025-<br>0716-<br>001 | IMG-<br>L1 ·<br>주간              | TRIM-<br>RH-001-<br>GY<br>트림 RH<br>회색 | 2,030     | FAB-GY-<br>C03                     | 12 roll         | 30<br>roll**      | 20<br>07<br>10        |

\* after WO-2025-0715-001 consumes 81 kg (projected)    \*\* after WO-2025-0715-002 consumes 12 roll (projected)

**i** **ETA Lead · 리드타임** — Plan Start - Lead (default 30 min). When ETA < CURRENT\_TIME, the row turns red and triggers immediate picking alert. Lead time configurable per line in MD-15 Line Master.

## 03 UI Component Specification

UI 컴포넌트 명세

### 3.1 Filter Toolbar · 필터

| ID     | COMPONENT                     | TYPE   | REQ | VALIDATION / BEHAVIOR                                                                                                                          | DEFAULT    |
|--------|-------------------------------|--------|-----|------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| CMP-01 | <b>Line Filter</b><br>라인 필터   | SELECT | N   | From <code>md_line</code> active rows. Selecting line filters all downstream lists.                                                            | All Lines  |
| CMP-02 | <b>Shift Filter</b><br>교대 필터  | SELECT | N   | 주간 (Day 08:00–20:00) / 야간 (Night 20:00–08:00).                                                                                                 | All Shifts |
| CMP-03 | <b>Status Filter</b><br>상태 필터 | SELECT | N   | Options: <code>Pending</code> / <code>Prep</code> / <code>Ready</code> / <code>Shortage</code> / <code>Released</code> . KPI cards auto-apply. | All Status |
| CMP-04 | <b>Date Picker</b><br>날짜 선택   | DATE   | N   | Filter by WO plan start date. Default today.                                                                                                   | Today      |
| CMP-05 | <b>Search Box</b><br>검색창      | TEXT   | N   | Match WO no., item code or item name. Min 2 chars. 300ms debounce.                                                                             | (empty)    |
| CMP-06 | <b>Excel Export</b><br>엑셀 출력  | BUTTON | N   | Filename: <code>WH06_Release_YYYYMMDD.xlsx</code> . Includes shortage column with red highlighting.                                            | —          |

### 3.2 KPI Strip · KPI

| ID     | METRIC                        | TYPE  | CALCULATION                                                                                                | REFRESH |
|--------|-------------------------------|-------|------------------------------------------------------------------------------------------------------------|---------|
| KPI-01 | <b>Today's WOs</b><br>금일 WO   | COUNT | <code>COUNT(*) FROM wo_header WHERE DATE(plan_start) = TODAY AND status IN ('Released', 'Active') .</code> | 5 min   |
| KPI-02 | <b>Ready to Pick</b><br>피킹 가능 | COUNT | WOs with all materials on-hand sufficient. Green.                                                          | 5 min   |

| ID     | METRIC                         | TYPE  | CALCULATION                                                                            | REFRESH |
|--------|--------------------------------|-------|----------------------------------------------------------------------------------------|---------|
| KPI-03 | <b>Within 30 min</b><br>30분 이내 | COUNT | WOs whose ETA (plan_start - lead) is within next 30 minutes. Yellow urgency.           | 1 min   |
| KPI-04 | <b>⚠ Shortage</b><br>자재 부족     | COUNT | WOs where any required material's on_hand < required (after upstream allocation). Red. | 5 min   |
| KPI-05 | <b>Released</b><br>출고 완료       | COUNT | WOs with all material release transactions complete today.                             | 5 min   |

### 3.3 Data Table Columns · 표 컬럼

| ID     | COLUMN                                 | SOURCE                     | FORMAT / RULE                                                                                           | SORT     |
|--------|----------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------|----------|
| COL-01 | <b>WO No.</b><br>작업지시                  | wo_header.wo_no            | Monospace. Click → PP-04 WO detail.                                                                     | ASC      |
| COL-02 | <b>Line · Shift</b><br>라인 · 교대         | wo_header.line_id          | Line code + shift name.                                                                                 | —        |
| COL-03 | <b>Finished Item</b><br>생산 품목          | md_item                    | EN main + KO subtext.                                                                                   | —        |
| COL-04 | <b>WO Qty</b><br>WO 수량                 | wo_header.plan_qty         | Right-align, thousands separator.                                                                       | —        |
| COL-05 | <b>Material</b><br>출고 자재               | wo_bom_demand.item_code    | Multiple rows if multiple materials per WO (one row per material). Monospace.                           | —        |
| COL-06 | <b>Required</b><br>소요량                 | wo_bom_demand.required_qty | WO qty × BOM unit ratio. Right-align with unit.                                                         | —        |
| COL-07 | <b>On Hand (projected)</b><br>현재고 (예상) | Calculated                 | Current on_hand minus upstream WO allocations. Red if insufficient. Tooltip shows allocation breakdown. | —        |
| COL-08 | <b>Plan Start</b><br>계획 시작             | wo_header.plan_start       | Time only if today; full datetime if other day.                                                         | ASC<br>★ |

| ID     | COLUMN             | SOURCE     | FORMAT / RULE                                                                              | SORT |
|--------|--------------------|------------|--------------------------------------------------------------------------------------------|------|
| COL-09 | ETA Lead<br>출고 ETA | Calculated | plan_start - line.lead_time .<br>Red if past, yellow if within 30min, default else.        | —    |
| COL-10 | Status<br>상태       | Derived    | Pending / Prep / Ready / Shortage / Released. Auto from rules.                             | —    |
| COL-11 | Action<br>액션       | Button     | [Start Picking] if Ready · [Prepare] if Prep · [Detail] if Pending · disabled if Shortage. | —    |

## 04 Event & Action Flow

이벤트 · 액션 흐름

1

**SCHEDULED** cron · WO demand refresh @ 5min

Every 5 minutes, system recalculates material demand for all active WOs by joining wo\_header × wo\_bom\_demand × wh\_inventory. Updates shortage flags.

5분마다 WO·BOM·재고를 조인하여 자재 소요·부족 플래그 재계산.

**EFFECT** Materialized view `vw_release_schedule` refreshed · WebSocket push to clients.

2

**USER** page.load · WH-06 진입

Warehouse staff opens screen. Default filter: today's date + all lines. Loads demand list sorted by ETA ascending.

WH-06 진입 시 기본 필터(오늘 + 전체 라인)로 ETA 오름차순 정렬 로딩.

**API** `GET /api/wh/release/schedule?date=today` .

3

**USER** click · [📄 Prepare]

User marks a WO as "preparing" — soft-reserves materials in inventory (status flag, not yet deducted). Other WOs see projected on-hand updated.

[Prepare] 클릭 시 자재를 소프트 예약(상태만 변경, 차감 아님). 다른 WO 는 예상 가용량 갱신 확인.

**API** `POST /api/wh/release/prepare` · `wo_header.prep_flag=1` · `reserved_qty` set.

**USER** click · [▶ Start Picking]

4

User initiates picking. System checks RBAC WH.Release and material availability. Creates picking session and launches WH-07 PDA flow.

[Start Picking] 클릭 시 권한 + 자재 가용성 확인 후 피킹 세션 생성, WH-07 PDA 흐름 시작.

**REDIRECT** POST /api/wh/release/start → returns  
picking\_id → redirect /wh/pda/release?pid=...

5

#### SYSTEM alert • ETA approaching

When ETA < 15 min and status is still Pending/Prep, system sends alert to warehouse staff and line supervisor.

ETA 15분 이내인데 Pending/Prep 상태인 경우 창고 담당·라인 슈퍼바이저에게 알림 발송.

**SIDE-EFFECT** In-app push notification · email · sys\_alert row inserted.

!

#### SYSTEM detect • shortage

When required material > projected on\_hand, WO turns red. [Start Picking] disabled. Suggestion: trigger MRP rerun or expedite PO from procurement.

소요량 > 예상 현재고 시 WO가 빨간색으로 전환되고 [Start Picking] 비활성. MRP 재계산·긴급 발주 권고.

**ALERT** WO marked shortage · email to procurement + plant mgr · ref to PP-05 MRP.

✓

#### SYSTEM return • from WH-07 success

After WH-07 picking completes, WO status = Released. Inventory deducted. Row badge updates. Production line notified that material is staged.

WH-07 피킹 완료 시 WO 상태 = Released. 재고 차감. 생산 라인에 자재 준비 완료 알림.

**DOWNSTREAM** WH-08 logs · WH-03 push update · PP-04 WO ready signal · INJ/IMG/PNT line dashboards.

## 05

## Exception & Error Handling

예외 · 에러 처리

E-WH06-001

### Material Shortage

자재 부족

**CAUSE** Required material qty > projected on-hand.  
소요량 > 예상 가용량.

**DETECT** 5-minute cron + real-time on WH-02/05/07 commits.

|          |                                                                                            |
|----------|--------------------------------------------------------------------------------------------|
| ACTION   | Row red. [Start Picking] disabled. Email procurement + plant mgr. Link to PP-05 MRP rerun. |
| RECOVERY | Expedite PO via WH-01 [Register Schedule] or transfer stock from another location.         |

🚫 SHORTAGE — EXPEDITE PROCUREMENT

E-WH06-002

### ETA Already Passed

ETA 경과

|          |                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------|
| CAUSE    | Current time > ETA but WO not yet picked.<br>현재 시각이 ETA를 초과했지만 미피킹.                                    |
| DETECT   | Per-row time check on every render.                                                                    |
| ACTION   | Row blinks red. ETA badge "OVERDUE". Auto-escalate notification every 5 min until picked or cancelled. |
| RECOVERY | Start picking immediately, or adjust WO plan_start in PP-04.                                           |

⚠ OVERDUE — START NOW

E-WH06-003

### Concurrent Picking Lock

중복 피킹 시도

|          |                                                                              |
|----------|------------------------------------------------------------------------------|
| CAUSE    | Another user has already started picking this WO.<br>동일 WO를 다른 사용자가 이미 시작.   |
| DETECT   | DB row lock on <code>picking_id</code> already exists.                       |
| ACTION   | Block. Show "Picking already in progress by [user] since [time]. View only?" |
| RECOVERY | Wait or coordinate with the other user.                                      |

🚫 LOCKED — IN PROGRESS

E-WH06-004

### WO Cancelled Upstream

WO 상위 취소

|          |                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | WO was cancelled in PP-04 after warehouse already prepared.<br>PP-04에서 WO 취소.                                                |
| DETECT   | WO status change event from PP module.                                                                                       |
| ACTION   | Row removed from list with notification: "WO cancelled. Return prepared materials." Soft-reservation released automatically. |
| RECOVERY | Return materials to storage. No inventory deduction needed (was only reserved).                                              |

⚠ WO CANCELLED — RETURN MATERIALS

E-WH06-005

### BOM Not Defined

BOM 미정의



|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| CAUSE    | WO created for an item with incomplete BOM in PP module.<br>PP BOM 미정의 품목 WO.       |
| DETECT   | LEFT JOIN <code>wo_bom_demand</code> returns 0 rows.                                |
| ACTION   | Material column shows "-". Row marked as "BOM ERROR" with link to MD-03 BOM Master. |
| RECOVERY | Master data team defines/completes BOM, then WH-06 row updates.                     |

⚠ NO BOM — SEE MD-03

E-WH06-006

### No Picker Assigned

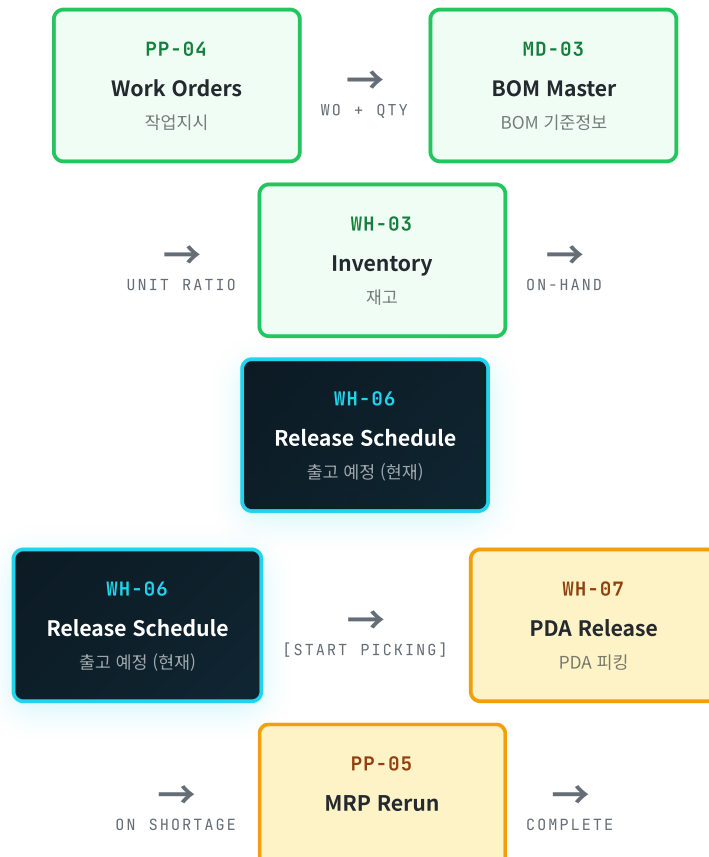
피커 미지정

|          |                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|
| CAUSE    | No warehouse staff with <code>WH.Release</code> role is online during shift.<br>교대 중 출고 권한 담당자 없음. |
| DETECT   | Hourly check during active shifts.                                                                 |
| ACTION   | Banner: "No picker available for [shift]." Alert sent to WH manager to assign or escalate.         |
| RECOVERY | Manager assigns staff or covers personally.                                                        |

i NO PICKER — ASSIGN STAFF

## 06 Linked Screens

선행 · 후행 연계 화면



MRP 재계산

WH-08

Transaction History

이력

## 07

### Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                                                                       | ACTION |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-04 | All outbound picking must follow FIFO (First In First Out) based on inbound date. WH-07 auto-suggests the oldest stock location.<br>출고 피킹은 입고일 기준 FIFO 자동 적용. WH-07이 최고령 재고 위치 자동 제안.                             | Auto   |
| BR-WH-13 | Picking cannot start if projected on-hand < required after upstream WO allocations. [Start Picking] button is disabled until materials are sufficient.<br>상위 WO 할당 후 예상 가용량이 소요량 미달이면 피킹 시작 불가. 자재 확보 시까지 버튼 비활성. | Block  |
| BR-WH-14 | Concurrent picking on the same WO is blocked by row lock. Only one active picking session per WO at any time.<br>동일 WO 동시 피킹 차단(행 잠금). 한 WO에 동시에 한 피킹 세션만 활성.                                                     | Block  |

A-MES

Index

WH WarehousePP PlanINJ InjectionIMG WrappingPNT PaintingQC QualityFG Finished GoodsMNT Main

L1A-MES Index > L2VOL.3 · WH Warehouse > L3WH-07 · PDA Release Picking

VOL.3 · WH MODULE

WAREHOUSE

창고 관리

L3 SCREENS · 화면 상세

WH-01Inbound Schedule  
입고 예정

WH-02PDA Inbound  
PDA 입고 처리

WH-03Inventory Status  
재고 현황

WH-04Location Map  
로케이션 맵

WH-05Inventory Adjust  
재고 조정

WH-06Release Schedule  
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WH-08Transaction History  
입출고 이력

ON THIS PAGE

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02 Screen Mockup

03 UI Components

04 Event & Action Flow

05 Exception Handling

06 Linked Screens

07 Business Rules

VOL.3 · WH · WAREHOUSE MANAGEMENT · PDA 출고 피킹

WH-07

PDA RELEASE PICKING

PDA 출고

DEVICE PDA

PHASE 1st Dev

ROLE Warehouse Staff (WH.Release)

FREQ Per WO · WO 단위

URL /wh/pda/reLease

01

Overview

화면 개요

PDA-guided FIFO picking workflow for material release to production lines. Three sequential steps: (1) **Scan WO barcode** to load picking list with system-suggested FIFO locations (oldest stock first), (2) **Scan item barcode at suggested location** — wrong items are immediately blocked, and (3) **Confirm picked quantity** via numeric keypad. Inventory is deducted only after final confirmation. The system enforces FIFO order strictly — if the worker tries a different location, a warning is shown with the correct location code.

생산 라인 자재 출고를 위한 PDA 기반 FIFO 피킹 흐름입니다. 3단계: (1) **WO 바코드 스캔**으로 시스템이 제안한 FIFO 위치 목록(최고령 재고 우선) 로드, (2) **제안 위치에서 품목 바코드 스캔** — 오폐품 즉시 차단, (3) **숫자패드로 피킹 수량 확인**. 최종 확정 후에만 재고 차감. FIFO 순서를 엄격히 강제하며, 다른 위치 시도 시 경고와 함께 올바른 위치 코드를 안내합니다.

|           |                                       |
|-----------|---------------------------------------|
| SCREEN ID | WH-07                                 |
| ENTRY     | From WH-06 [Start Picking] 출고 예정에서 진입 |
| LAYOUT    | 3-step PDA wizard 3단계 PDA 위저드         |

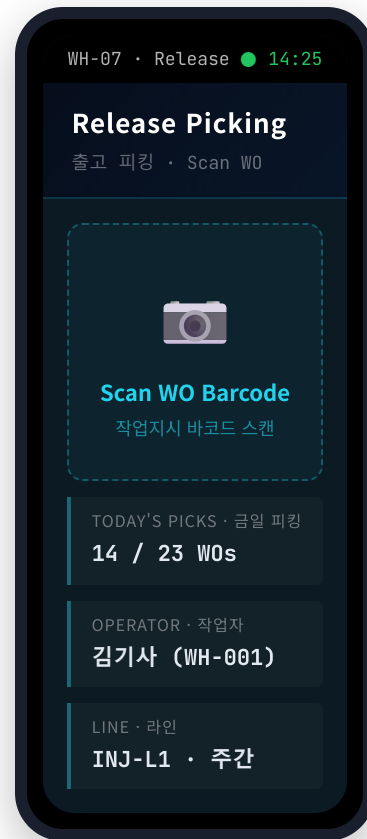
|          |                                           |
|----------|-------------------------------------------|
| HARDWARE | Zebra TC22 · 1D/2D scanner                |
| RULE     | FIFO enforced (BR-WH-04) FIFO 강제          |
| TABLES   | wh_release , wh_inventory , wo_bom_demand |
| RBAC     | WH.Release                                |

## 02 Screen Mockup

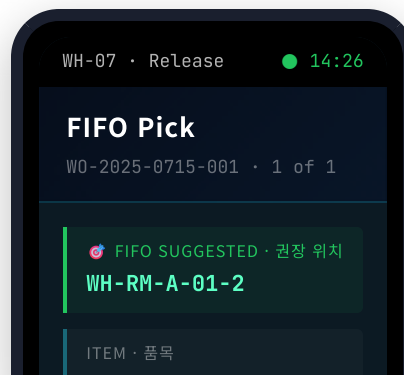
화면 목업 — PDA 3단계 + 에어 케이스

### ● WH-07 · PDA RELEASE PICKING · 3-STEP WIZARD

#### ① SCAN WO · 작업지시 스캔



#### ② SCAN ITEM @ LOCATION · 위치·품목 스캔



RM-PP-NAT-20

FIFO DATE · 입고일  
2025-06-28 09:14

REQUIRED · 필요  
81 kg

  
Scan Item Barcode  
품목 바코드 스캔

③ CONFIRM QTY · 수량 확정

WH-07 · Release ● 14:27

Confirm Pick Qty

RM-PP-NAT-20 ✓

ITEM VERIFIED ✓  
RM-PP-NAT-20

LOCATION · 위치  
WH-RM-A-01-2

REQUIRED · 필요  
81 kg

81

123

456

789

All0✕

✓ Confirm Release

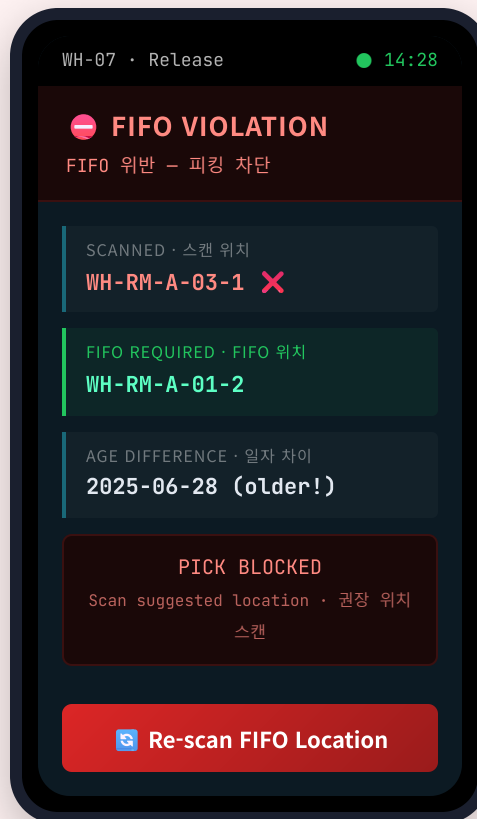
🚫 Error Case · 에러 케이스 — Wrong Item / Wrong Location · 오폭목 · 잘못된 위치

**Trigger · 발생 시점**  
— Step ②  
스캔한 위치가 FIFO 권장 위치와 다른 경우.

**Action · 동작** — 즉시 차단 + 빨간 화면 + 진동 + 비프음. 권장 위치 + 입고일자 차이 명확히 표시.

**Override 옵션** — 슈퍼바이저 PIN 입력 시 1회 예외 허용 (감사 로그 기록, RPT-04에서 통계 추적).

**Other errors · 기타 에러** — Wrong item scan, qty > on\_hand, location empty, network offline.



### 3.1 Step ① — WO Scan · 작업지시 스캔

| ID     | COMPONENT                               | TYPE   | REQ | VALIDATION / BEHAVIOR                                                                                         | ON SUCCESS |
|--------|-----------------------------------------|--------|-----|---------------------------------------------------------------------------------------------------------------|------------|
| CMP-01 | <b>WO Barcode Scanner</b><br>WO 바코드 스캐너 | SCAN   | Y   | Reads WO no. Validates against picking session (must match <code>picking_id</code> launched from WH-06).      | → Step ②   |
| CMP-02 | <b>Picking List Display</b><br>피킹 목록 표시 | LIST   | —   | Shows N of M materials, each with FIFO-suggested location. Ordered by location proximity for efficient route. | —          |
| CMP-03 | <b>Manual Entry</b><br>수동 입력            | BUTTON | N   | Fallback if WO label damaged. Keyboard input. Same validation. Logs <code>manual=true</code> .                | → Step ②   |

### 3.2 Step ② — Location & Item Scan · 위치 · 품목 스캔

| ID     | COMPONENT                                   | TYPE  | REQ | VALIDATION / BEHAVIOR                                                                                          | ON SUCCESS  |
|--------|---------------------------------------------|-------|-----|----------------------------------------------------------------------------------------------------------------|-------------|
| CMP-04 | <b>FIFO Suggested Display</b><br>FIFO 권장 위치 | LABEL | —   | Shows the location with oldest <code>inbound_date</code> for the required item. Green highlight + map preview. | —           |
| CMP-05 | <b>Location QR Scanner</b><br>위치 QR 스캐너     | SCAN  | Y   | Must match suggested location (BR-WH-04). Override allowed only with supervisor PIN.                           | → item scan |
| CMP-06 | <b>Item Barcode Scanner</b>                 | SCAN  | Y   | Verifies scanned item matches                                                                                  | → Step ③    |

| ID     | COMPONENT                              | TYPE | REQ | VALIDATION / BEHAVIOR                                                                         | ON SUCCESS |
|--------|----------------------------------------|------|-----|-----------------------------------------------------------------------------------------------|------------|
|        | 품목 바코드 스캐너                             |      |     | required<br><code>item_code</code> .<br>Wrong item → block (E-WH07-002).                      |            |
| CMP-07 | <b>Supervisor Override</b><br>슈퍼바이저 예외 | PIN  | N   | 6-digit PIN.<br>Allows FIFO violation for one pick.<br>Logged with reason text and timestamp. | → Step ②   |

### 3.3 Step ③ — Quantity Confirmation · 수량 확정

| ID     | COMPONENT                              | TYPE    | REQ | VALIDATION / BEHAVIOR                                                                                                                       | ON SUCCESS |
|--------|----------------------------------------|---------|-----|---------------------------------------------------------------------------------------------------------------------------------------------|------------|
| CMP-08 | <b>Numeric Keypad</b><br>숫자패드          | NUMPAD  | Y   | 0-9, decimal. Max 6 digits + 3 decimals. Must be ≤ on_hand at the location.                                                                 | → Confirm  |
| CMP-09 | <b>[All Required] Button</b><br>필요량 전체 | BUTTON  | N   | Auto-fills with required qty from BOM demand. Common case shortcut.                                                                         | → Confirm  |
| CMP-10 | <b>Qty Display</b><br>수량 디스플레이         | DISPLAY | —   | Large monospace digits. Green if = required, yellow if < required (partial), red if > on_hand.                                              | —          |
| CMP-11 | <b>Partial Pick Toggle</b><br>부분 피킹    | AUTO    | —   | If qty < required, system asks "Continue to next location for remaining?" and re-routes to next FIFO location.                              | —          |
| CMP-12 | <b>[Confirm Release]</b><br>출고 확정      | BUTTON  | Y   | Commits atomically: deduct <code>wh_inventory</code> , insert <code>wh_release</code> , update WO status, log <code>wh_transaction</code> . | Complete   |



1

**USER** launch · from WH-06 [Start Picking]

PDA launches with picking\_id context. Loads WO + material list + FIFO-suggested locations.

WH-06 [Start Picking] 클릭으로 PDA 실행. WO·자재 목록·FIFO 권장 위치 로드.

**CONTEXT** `picking_id`, `wo_no`, materials[] (with FIFO locations).

2

**PDA SCAN** trigger · WO barcode

Worker scans WO barcode label on the work order travel sheet to confirm picking session.

작업자가 워크오더 종이의 WO 바코드를 스캔하여 피킹 세션 확정.

**VALIDATE** Scanned WO == session WO · else block.

3

**SYSTEM** compute · FIFO route

System calculates pick route: for each required material, finds locations with stock and orders by inbound\_date ASC (oldest first). Displays first FIFO location.

자재별로 재고 보유 위치를 입고일 오름차순(최고령 우선)으로 정렬하여 피킹 경로 산출. 첫 FIFO 위치 표시.

**QUERY** `SELECT location, on_hand, MIN(inbound_date) FROM wh_inventory ORDER BY inbound_date ASC`.

4

**PDA SCAN** trigger · location QR

Worker walks to suggested location and scans the shelf QR code. System validates it matches the FIFO suggestion.

작업자가 권장 위치로 이동 후 선반 QR 스캔. 시스템이 FIFO 일치 여부 검증.

**VALIDATE** Scanned == suggested · else E-WH07-001 (FIFO violation).

5

**PDA SCAN** trigger · item barcode

Worker scans the item barcode at the location.

System verifies it matches the required item code for this WO.

작업자가 위치의 품목 바코드 스캔. 시스템이 WO 필요 품목과 일치 여부 검증.

```
VALIDATE scanned == required_item · else E-WH07-002 (wrong item).
```

6

**USER** tap · numpad / [All Required]

Worker enters picked qty. System checks vs on\_hand at location and vs required qty.

수량 입력. 위치 현재고 및 필요량과 대조.

```
CHECK qty ≤ location.on_hand · if < required, partial mode → re-route to next FIFO location.
```

7

**USER** tap · [Confirm Release]

Worker confirms. Backend commits transaction atomically: deduct inventory at location, insert release row, log transaction, increment WO picked qty.

확정. 백엔드 트랜잭션 원자적 커밋: 재고 차감, 출고 행 INSERT, 거래 로그, WO 피킹량 증가.

```
API POST /api/wh/release/confirm · BEGIN TRAN ... COMMIT .
```

✓

**SYSTEM** complete · loop or finish

If more materials in picking list, loop to step 4 with next material. When all picked, WO status = Released, line dashboard notified, return to WH-06.

자재가 더 있으면 step 4로 루프. 전부 완료 시 WO Released 처리, 라인 대시보드 알림, WH-06 복귀.

```
DOWNSTREAM PP-04 WO ready signal · INJ/IMG/PNT line · WH-03/04 update · WH-08 log.
```

## 05

### Exception & Error Handling

예외 · 에러 처리

**E-WH07-001**

#### FIFO Violation

FIFO 위반

CAUSE

Worker scans location with newer stock instead of FIFO-suggested oldest.

FIFO 권장 위치가 아닌 다른 위치 스캔.

DETECT

Step ② location scan comparison.

**ACTION** Red full-screen + vibration + beep. Show suggested location + age difference. [Re-scan] button.

**RECOVERY** Re-scan correct FIFO location · OR supervisor PIN override (audit logged).

— FIFO VIOLATION — SCAN OLDEST

E-WH07-002

### Wrong Item Scanned

오폐목 스캔

**CAUSE** Scanned item ≠ required item for this WO.  
스캔 품목이 WO 필요 품목과 다름.

**DETECT** Step ② item scan validation.

**ACTION** Block. Red screen + beep + vibration. Required item code shown prominently.

**RECOVERY** Re-scan correct item. If physical mislabel suspected, escalate to QC-04.

— WRONG ITEM — RE-SCAN

E-WH07-003

### Insufficient at Location

위치 재고 부족

**CAUSE** Location on\_hand < required (other WO already consumed).  
다른 WO가 먼저 소비하여 부족.

**DETECT** Step ③ qty entry validation.

**ACTION** Allow partial pick of available qty. Auto-route to next FIFO location for remainder.

**RECOVERY** Continue picking at suggested next location.

△ PARTIAL — CONTINUE NEXT LOC

E-WH07-004

### Supervisor Override

슈퍼바이저 예외

**CAUSE** FIFO violation accepted via supervisor PIN.  
슈퍼바이저 PIN으로 FIFO 예외.

**DETECT** Override applied at Step ② re-scan.

**ACTION** Allow pick to proceed. Mandatory reason text input. Logged with supervisor ID, timestamp, original/actual location.

**RECOVERY** N/A. Audit reviewed weekly via RPT-04.

△ OVERRIDE LOGGED — AUDIT

**E-WH07-005****Network Disconnected**

네트워크 단절

|          |                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------|
| CAUSE    | WiFi signal lost during picking.<br>피킹 중 WiFi 단절.                                                       |
| DETECT   | Heartbeat timeout 30s.                                                                                  |
| ACTION   | Queue confirmation locally (4 hr buffer). Yellow banner. Pick can continue offline; sync on reconnect.  |
| RECOVERY | On reconnect, queued releases commit in FIFO order. Conflict resolution if inventory changed meanwhile. |

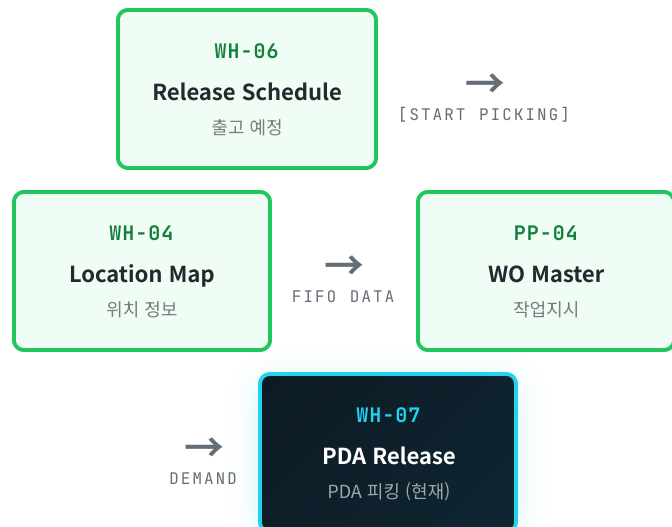
**⚠ OFFLINE — QUEUE MODE****E-WH07-006****WO Cancelled During Picking**

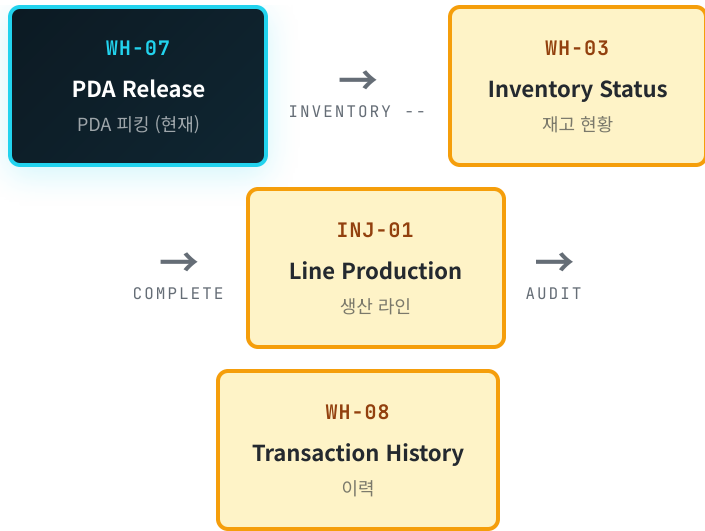
피킹 중 WO 취소

|          |                                                                                                                                |
|----------|--------------------------------------------------------------------------------------------------------------------------------|
| CAUSE    | PP-04 cancels WO while worker is picking.<br>피킹 진행 중 WO 취소.                                                                    |
| DETECT   | WO status push event from PP module.                                                                                           |
| ACTION   | Pause picking. Modal: "WO cancelled. Return items already picked to original locations?" Auto-create return adjustment ticket. |
| RECOVERY | Worker returns items, system creates compensating release-back transactions.                                                   |

**⚠ WO CANCELLED — RETURN ITEMS****06****Linked Screens**

선행 · 후행 연계 화면





## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                                                                                                          | ACTION          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| BR-WH-04 | <p>All outbound picking must follow FIFO based on inbound date. System auto-suggests oldest stock location; deviations require supervisor PIN override with mandatory reason.</p> <p>출고 피킹은 입고일 기준 FIFO 자동 적용. 권장 위치 외 피킹은 슈퍼바이저 PIN + 사유 입력 필수.</p> | Auto + Override |
| BR-WH-15 | <p>Items scanned during picking must match the required item_code for the WO. Wrong item is immediately blocked with no override option.</p> <p>피킹 중 스캔 품목은 WO 필요 품목과 일치해야 함. 오폐품 즉시 차단 (예외 없음).</p>                                                 | Block           |
| BR-WH-16 | <p>FIFO override events are logged and reviewed weekly via RPT-04. More than 5 overrides in 30 days by a single supervisor triggers compliance review.</p> <p>FIFO 예외 이벤트는 RPT-04에서 주간 검토. 30일 내 1인 5건 초과 시 컴플라이언스 리뷰 트리거.</p>                       | Audit           |

A-MES · VOL.3 · WH-07 PDA Release  
Picking · L3 Detail Spec v1.0 ·  
Bilingual EN/KO

Automotive Parts MES ·  
SAV & GEO Supply ·  
Confidential

A-MES

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입출고 이력

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입출고이력

DEVICE

Web PC

PHASE

1st Dev

ROLE

Manager · Auditor

FREQ

On-demand · 감사시

URL

/wh/transactions

01

Overview

화면 개요

Complete audit trail of every inventory movement: **IN** (WH-02 inbound), **OUT** (WH-07 release), **ADJ+/ADJ-** (WH-05 adjustment), **RELOC** (location move). Each row captures: timestamp, operator, transaction type, item, location, qty before/after, reason code, reference document (PO / WO / Adjustment ID), and approver (for adjustments). Searchable by date range, item, operator, transaction type, and reference ID. Excel export for audit and regulatory purposes. Records are immutable and retained for **10 years** per quality system requirements.

모든 재고 이동에 대한 완전한 감사 기록: **입고(IN, WH-02)**, **출고(OUT, WH-07)**, **조정(ADJ+/ADJ-, WH-05)**, **위치 이동(RELOC)**. 각 행에는 일시·작업자·거래 유형·품목·위치·전후 수량·사유 코드·참조 문서(PO/WO/Adjustment ID)·승인자(조정의 경우)가 기록됩니다. 날짜 범위·품목·작업자·거래 유형·참조 ID로 검색 가능하며, 감사·규제 목적의 Excel 출력을 지원합니다. 기록은 **수정 불가**하며 품질 시스템 요건에 따라 **10년 보관**됩니다.

|           |                                             |
|-----------|---------------------------------------------|
| SCREEN ID | WH-08                                       |
| LAYOUT    | Filter bar + KPI + Audit table   필터·KPI·감사표 |
| REFRESH   | Real-time append (WebSocket)   실시간 추가       |
| RETENTION | <b>10 years</b> · immutable   10년 보관·수정 불가  |
| TABLES    | wh_transaction   (append-only)              |
| RBAC      | WH.Audit   · export: WH.Export              |
| PHASE     | 1st Dev                                     |

## 02 Screen Mockup

화면 목업

● WH-08 · TRANSACTION HISTORY · WEB PC · 1920×1080

## A-MES · Warehouse · Transaction History

Warehouse &gt; History

2025-07-15



~

2025-07-15



All Types



Item / Operator / Ref ID...

All Operators



Search

Excel Export

Audit PDF

28

IN · 입고  
금일 입고

42

OUT · 출고  
금일 출고

3

ADJ · 조정  
금일 조정

1

RELOC · 이동  
위치 이동

74

Total Txns  
총 거래

| Timestamp<br>일시        | Type<br>유형 | Item<br>Code<br>품목       | Location<br>위치                               | Qty<br>Before | Δ   | Qty<br>After | Reas<br>Ref<br>사유                                |
|------------------------|------------|--------------------------|----------------------------------------------|---------------|-----|--------------|--------------------------------------------------|
| 2025-07-15<br>09:14:23 | IN         | FAB-<br>GY-<br>C03       | WH-FAB-<br>D-01-1                            | 22            | +20 | 42           | PO-20<br>0701-<br>현대원<br>주                       |
| 2025-07-15<br>14:27:08 | OUT        | RM-<br>PP-<br>NAT-<br>20 | WH-RM-<br>A-01-2                             | 1,250         | -81 | 1,169        | WO-2<br>0715-<br>INJ-L                           |
| 2025-07-15<br>15:32:11 | ADJ-       | FAB-<br>GY-<br>C03       | WH-FAB-<br>D-01-1                            | 42            | -2  | 40           | COUN<br>ADJ-2<br>0715                            |
| 2025-07-15<br>16:05:42 | IN         | PNT-<br>BK-<br>NH578     | WH-PNT-<br>E-01-1                            | 234           | +50 | 284          | PO-20<br>0705-<br>KCC5                           |
| 2025-07-15<br>17:42:33 | ADJ-       | RM-<br>PP-<br>NAT-<br>20 | WH-RM-<br>A-01-2                             | 1,169         | -45 | 1,124        | DAMA<br>ADJ-2<br>0715<br>지게차                     |
| 2025-07-15<br>18:30:17 | RELOC      | PNT-<br>BK-<br>NH578     | WH-PNT-<br>E-01-1<br>→ WH-<br>PNT-E-03-<br>1 | 284 /<br>0    | ±3  | 281 /<br>3   | RELO<br>ADJ-2<br>0715                            |
| 2025-07-15<br>19:14:55 | OUT        | FAB-<br>GY-<br>C03       | WH-FAB-<br>D-01-1                            | 40            | -12 | 28           | WO-2<br>0715-<br>IMG-1<br>· FIFO<br>overr<br>일 신 |

Showing 1-7 of 74 transactions · 1-7  
/ 74 거래

&lt; Prev 1 2 3 ... 11 Next &gt;



★ **Immutable audit log · 수정 불가 감사 기록** — 본 화면의 모든 행은 INSERT-only 테이블에 기록되어 수정·삭제가 불가합니다. 변경이 필요할 경우 보정 거래(compensating transaction)를 별도 행으로 기록합니다. 행 색상: 노랑(FIFO Override), 빨강(고가 Damage), 일반.

## 03 UI Component Specification

UI 컴포넌트 명세

### 3.1 Filter Toolbar · 필터

| ID     | COMPONENT                 | TYPE     | REQ | VALIDATION / BEHAVIOR                                                                                                     | DEFAULT   |
|--------|---------------------------|----------|-----|---------------------------------------------------------------------------------------------------------------------------|-----------|
| CMP-01 | Date Range<br>기간          | DATE × 2 | Y   | From ≤ To. Max range 90 days. Beyond 90 days requires <code>WH.Audit.Extended</code> role.                                | Today     |
| CMP-02 | Type Filter<br>유형 필터      | SELECT   | N   | Options: <code>ALL</code> / <code>IN</code> / <code>OUT</code> / <code>ADJ</code> / <code>RELOC</code> . Multi-select v2. | All Types |
| CMP-03 | Search Box<br>검색창         | TEXT     | N   | Search by item code/name, location code, or reference ID (PO/WO/ADJ). 300ms debounce.                                     | (empty)   |
| CMP-04 | Operator Filter<br>작업자 필터 | SELECT   | N   | From <code>md_user</code> where role includes WH.* roles. Searchable dropdown.                                            | All       |
| CMP-05 | Excel Export<br>엑셀 출력     | BUTTON   | N   | Requires <code>WH.Export</code> role. Max 50,000 rows. Filename: <code>WH08_Txn_YYYYMMDD_YYYYMMDD.xlsx</code> .           | —         |
| CMP-06 | Audit PDF<br>감사 PDF       | BUTTON   | N   | Generates PDF with cover page (period, filters, generator name) and audit footer on every page. Digitally signed.         | —         |

### 3.2 KPI Strip · KPI

| ID     | METRIC         | TYPE  | CALCULATION                                           | FILTER ON CLICK         |
|--------|----------------|-------|-------------------------------------------------------|-------------------------|
| KPI-01 | IN<br>입고       | COUNT | COUNT WHERE <code>type='IN'</code> within date range. | <code>type=IN</code>    |
| KPI-02 | OUT<br>출고      | COUNT | COUNT WHERE <code>type='OUT'</code> .                 | <code>type=OUT</code>   |
| KPI-03 | ADJ<br>조정      | COUNT | COUNT WHERE <code>type IN ('ADJ+', 'ADJ-')</code> .   | <code>type=ADJ</code>   |
| KPI-04 | RELOC<br>위치 이동 | COUNT | COUNT WHERE <code>type='RELOC'</code> .               | <code>type=RELOC</code> |

| ID     | METRIC             | TYPE  | CALCULATION               | FILTER ON CLICK |
|--------|--------------------|-------|---------------------------|-----------------|
|        |                    |       | Paired entries (– and +). |                 |
| KPI-05 | Total Txns<br>총 거래 | COUNT | Sum of all types.         | All             |

### 3.3 Table Columns · 표 컬럼

| ID     | COLUMN                   | SOURCE                                 | FORMAT / RULE                                                                          | SORT      |
|--------|--------------------------|----------------------------------------|----------------------------------------------------------------------------------------|-----------|
| COL-01 | Timestamp<br>일시          | wh_transaction.txn_time                | Format: YYYY-MM-DD<br>HH:MM:SS . KST<br>timezone.<br>Monospace.                        | DESC<br>★ |
| COL-02 | Type<br>유형               | wh_transaction.type                    | Badge: IN (green) /<br>OUT (red) / ADJ+/-<br>(yellow/red) / RELOC<br>(purple).         | –         |
| COL-03 | Item Code<br>품목          | md_item                                | Monospace. Click →<br>MD-04 item master.                                               | –         |
| COL-04 | Location<br>위치           | md_location                            | For RELOC: shows<br>from → to. Click →<br>WH-04 map.                                   | –         |
| COL-05 | Qty Before<br>전 수량       | wh_transaction.qty_before              | Right-align.<br>Numerical snapshot<br>at time of<br>transaction.                       | –         |
| COL-06 | Δ<br>증감                  | wh_transaction.delta                   | Signed. Green<br>positive, red<br>negative, purple<br>paired (RELOC).                  | –         |
| COL-07 | Qty After<br>후 수량        | wh_transaction.qty_after               | Right-align, bold. =<br>qty_before + delta.                                            | –         |
| COL-08 | Reason /<br>Ref<br>사유·참조 | wh_transaction.reason_code<br>+ ref_id | Reason code (bold) +<br>ref doc (PO/WO/ADJ<br>ID). Click ref →<br>opens source screen. | –         |
| COL-09 | Operator<br>작업자          | md_user.name                           | User who performed<br>transaction. Click →<br>user profile.                            | –         |
| COL-10 | Approver<br>승인자          | wh_transaction.approver_id             | Populated for<br>ADJ/RELOC/Override.<br>"—" for routine<br>IN/OUT.                     | –         |

1

**SYSTEM trigger · transaction commit**

Whenever WH-02/05/07 commits an inventory change, a corresponding row is written to wh\_transaction via DB trigger. Atomic with source operation.

WH-02-05-07이 재고 변경 커밋 시, DB 트리거로 wh\_transaction에 행이 자동 기록됨. 원자적 커밋.

**EFFECT** `INSERT INTO wh_transaction (...) VALUES (...)` · WebSocket push.

2

**USER page.load · WH-08 진입**

Auditor or warehouse manager opens screen. Default filter: today, all types. Loads first page of 50 rows sorted by timestamp DESC.

감사자 또는 관리자가 진입. 기본 필터(오늘 · 전체 유형)로 첫 50행 로딩(시각 내림차순).

**API** `GET /api/wh/transactions?date=today&page=1&size=50` ·

3

**SYSTEM push · new transaction**

When a new transaction commits, WebSocket pushes the row to all subscribed clients. New row appears at top with brief flash animation.

신규 거래 커밋 시 WebSocket으로 푸시. 표 상단에 새 행이 짧은 플래시 애니메이션과 함께 추가.

**UI** Row prepended · auto-pagination preserved · KPI counters increment.

4

**USER click · KPI card**

User clicks a KPI to drill into that type. Filter applied, table redraws.

KPI 클릭 시 해당 유형 필터 자동 적용 후 표 갱신.

**URL** `?type=ADJ&date=...` · deep-linkable.

5

**USER click · ref\_id link**

User clicks a reference (PO / WO / ADJ ID) to navigate to source document.

참조 ID(PO/WO/ADJ) 클릭 시 원본 문서 화면으로 이동.

**NAV** PO → WH-01 detail · WO → PP-04 · ADJ → WH-05.

6

USER click · [📄 Excel Export]

User requests export. System checks WH.Export role and row count. If < 50,000, generates .xlsx async via background job; download link emailed when ready.

Excel 출력 요청. 권한·행 수 검증 후 비동기 백그라운드 작업으로 생성. 완료 시 다운로드 링크 이메일 전송.

API POST /api/wh/export · job\_id returned · email when ready.

✓

USER click · [📄 Audit PDF]

Generates a digitally signed PDF report with cover page, audit footer, and tamper-evident hash. Used for ISO/IATF audit submission.

디지털 서명된 PDF 감사 보고서 생성(표지·푸터·해시값 포함). ISO/IATF 감사 제출용.

OUTPUT PDF with SHA-256 hash footer · timestamped digital signature.

## 05

## Exception & Error Handling

예외 · 에러 처리

E-WH08-001

### Range Exceeds 90 Days

기간 90일 초과

|          |                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------|
| CAUSE    | Date range > 90 days. Performance and storage concern.<br>조회 기간 90일 초과.                               |
| DETECT   | Client + server validation.                                                                           |
| ACTION   | Block. Unless user has WH.Audit.Extended role. Modal: "Range > 90 days requires extended audit role." |
| RECOVERY | Reduce range or request extended role from SYS admin.                                                 |

⚠ RANGE TOO LARGE — REDUCE OR ESCALATE

E-WH08-002

### Edit Attempt

수정 시도

|          |                                                                                              |
|----------|----------------------------------------------------------------------------------------------|
| CAUSE    | User attempts to edit/delete a transaction row via API or DevTools.<br>감사 행 수정·삭제 시도.        |
| DETECT   | UPDATE/DELETE on wh_transaction blocked at DB level (BR-WH-17).                              |
| ACTION   | Return 403. Log attempt with user_id, IP, timestamp in sys_security_audit . Alert SYS admin. |
| RECOVERY | If correction needed, create compensating transaction via WH-05 with reason SYSTEM_ANOMALY . |

🚫 IMMUTABLE — CREATE COMPENSATION

E-WH08-003

### Export Row Limit

출력 행 제한

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| CAUSE    | Filtered result > 50,000 rows.<br>결과 5만 행 초과.                                   |
| DETECT   | Server-side count check.                                                        |
| ACTION   | Reject. Suggest narrower filter or use RPT-04 audit reports for large datasets. |
| RECOVERY | Narrow filter, or request RPT-04 audit batch report.                            |

⚠ TOO MANY ROWS — USE RPT-04

E-WH08-004

### Hash Mismatch

해시 불일치

|          |                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------|
| CAUSE    | Periodic integrity check detects record hash mismatch (data tampered or corruption).<br>레코드 해시 불일치(변조·손상). |
| DETECT   | Daily integrity cron at 02:00 KST.                                                                         |
| ACTION   | CRITICAL alert to SYS admin, plant director, QA lead. Affected rows flagged. Forensic backup loaded.       |
| RECOVERY | SYS team investigates; restore from immutable archive; report to compliance.                               |

🛑 INTEGRITY VIOLATION — INVESTIGATE

E-WH08-005

### Old Data Archived

오래된 데이터 아카이브

|          |                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|
| CAUSE    | Searching for data older than 2 years; moved to cold storage.<br>2년 초과 데이터는 콜드 스토리지.               |
| DETECT   | Query response time slow (> 5s).                                                                   |
| ACTION   | Show banner: "Querying archived data. May take longer." Background async fetch + email when ready. |
| RECOVERY | Wait for fetch or use specific filters to narrow.                                                  |

📁 ARCHIVED — FETCHING ASYNC

E-WH08-006

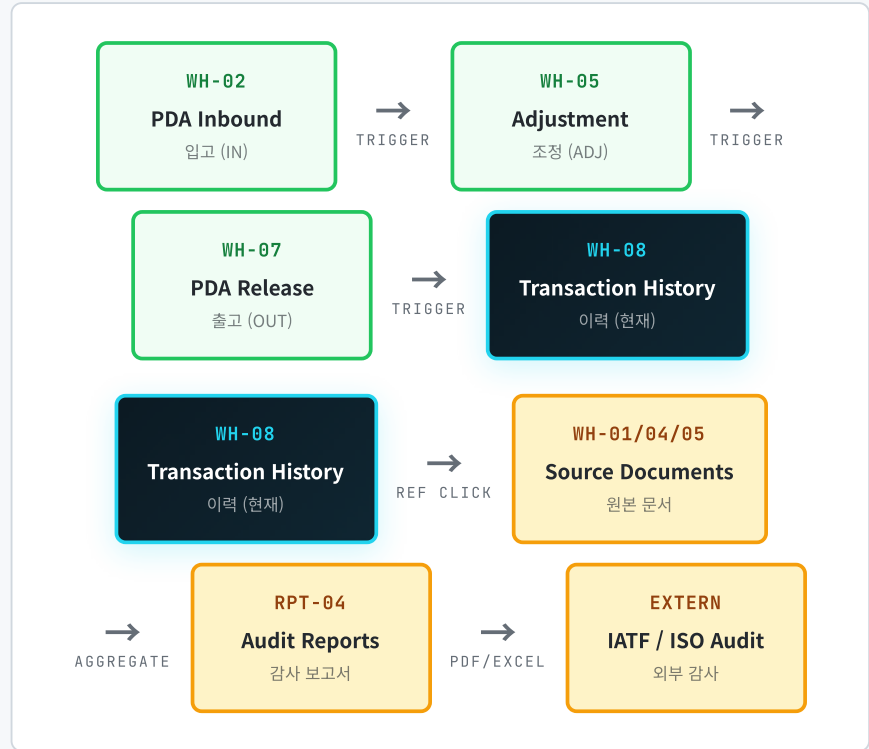
### No Transactions Found

거래 없음

|          |                                                                               |
|----------|-------------------------------------------------------------------------------|
| CAUSE    | Filter combination yields zero results.<br>필터 결과 0건.                          |
| DETECT   | API returns empty array.                                                      |
| ACTION   | Empty state illustration: "No transactions match your filters." Reset button. |
| RECOVERY | Reset or adjust filters.                                                      |

## 06 Linked Screens

선행 · 후행 연계 화면



## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID  | DESCRIPTION                                                                                                                                                                                                                           | ACTION |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BR-WH-17 | All transactions in wh_transaction are immutable (INSERT-only). UPDATE and DELETE are forbidden at DB level. Corrections must use compensating transactions.<br>모든 거래는 수정 불가(INSERT 전용). UPDATE·DELETE는 DB 차원에서 금지. 보정은 별도 보정 거래로 처리. | Block  |
| BR-WH-18 | Transaction records are retained for 10 years to comply with IATF 16949 traceability requirements. Records older than 2 years move to cold storage.<br>IATF 16949 추적 요건에 따라 거래 기록 10년 보관. 2년 초과 데이터는 콜드 스토리지 이관.                      | Auto   |

| RULE ID  | DESCRIPTION                                                                                                                                                                                                         | ACTION    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| BR-WH-19 | <p>Each transaction row includes a SHA-256 hash of its content. Daily integrity check verifies no tampering. Hash mismatch = CRITICAL incident.</p> <p>각 거래 행에 SHA-256 해시 포함. 일일 무결성 검증으로 변조 감지. 불일치는 치명적 인시던트.</p> | Integrity |